

HWO ExoEarth Yields for a Mass-Selected Planet Sample in a Narrowed Habitable Zone

A rederivation of the AYO yield model of Stark et al., laid out as a probabilistic model

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Abstract

How many exoEarth candidates (EECs) would the baseline Habitable Worlds Observatory (HWO) mission of Stark et al. (2024) find if the definition of an exoEarth were changed? Stark’s canonical definition counts planets smaller than $1.4 R_{\oplus}$ in a habitable zone of 0.95–1.67 AU, scaled by $\sqrt{L_{\star}}$. We instead count planets of $0.5\text{--}2 M_{\oplus}$ (with $R = M^{0.27}$) in a narrower zone of 0.96–1.20 AU, scaled the same way. Because the survey redistributes its observing time whenever the number of planets it expects to find changes, the answer cannot be obtained by rescaling Stark’s yield; the survey optimization must be run again.

We rederive the Altruistic Yield Optimization (AYO) model from its published equations and parameters, follow Stark’s documented methods wherever they are stated, and write the model as a probabilistic graphical model whose only decision is how the survey allocates its time. Nothing is fitted to Stark’s results. The model’s expected 6 m yield is 20.92, against a published 22.5. The median canonical yield is 18.8 (90% interval 5.6–53.4), and the median redefined yield is 4.3 (1.2–14.8). Their ratio, $0.232^{+0.027}_{-0.021}$ (68% interval), exceeds the occurrence ratio of 0.204 because the narrower zone discards the faintest planets, and it stays between 0.2201 and 0.2386 across seven model variants. The chance of finding at least 25 candidates falls from 35.1% to 1.20%.

Of 24 independent comparisons with the published record, 21 agree. Two disagreements remain, and each reduces to a question that only the AYO authors can answer. First, the time needed to take a planet’s spectrum depends more steeply on target difficulty and on aperture than in Stark’s model, and a 56-point study shows that no setting of the throughput parameters removes the difference. Second, the published uncertainty interval on $\eta_{\oplus}$ admits two readings, and that single choice accounts for 99.1% of the remaining disagreement with Stark’s yield distribution. Stark’s own albedo test, adopted here, gives an albedo penalty of 3.6% rather than the 12% he reports.

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1 Introduction

Yield estimates for HWO rest on the AYO framework (Stark et al. 2014, 2019, 2024). AYO allocates exposure time across a target list by an equal-slope rule, so its output responds to a change in the planet population in ways a proportionality argument does not capture: when fewer detections are expected, characterization time is freed and the survey reallocates it. A question about a different planet population therefore requires the survey to be re-optimized.

This report answers one such question. We ask how many planets the baseline HWO mission would find around FGK stars if

- planets were selected by mass, $0.5\text{--}2 M_{\oplus}$ with $R = M^{0.27}$ (Chen & Kipping 2017), and
- the habitable zone were $0.96\text{--}1.20$ AU for the Sun, scaled as $\sqrt{L_{\star}}$ for other stars.

Throughout, Stark’s selection is called the *canonical box* and ours the *redefined box*. AYO’s code is not public, so we reimplement the model from the published equations and parameter tables (Secs. 2 and 3), validate it against every published number we can reach (Sec. 4), and then evaluate both boxes (Sec. 5). Section 6 measures how much the answer depends on settings that the published record leaves open, and Sec. 7 describes the one discrepancy we cannot remove.

2 The model

2.1 The model as an influence diagram

AYO is usually described as a procedure: compute completeness, optimize exposure times, count planets, then add sources of uncertainty one at a time. That description hides the structure that determines the result: which quantities are random and which are fixed, and which random quantities the survey planner can see when it decides how to spend its time. Stark et al. (2024, Sec. 3) call the last distinction *actionable* versus *static* uncertainty. A graphical model makes it precise.

Figure 1 shows the model as an influence diagram: a directed acyclic graph of random variables with a single *decision* node, the survey allocation. Table 1 describes each node and the pipeline step that implements it.

Node	Type	Specification	Source; step
$\eta_{\oplus}$	random, global	lognormal, median 0.257, ln-width 0.76 (Sec. 3.2)	Stark+2024 Sec. 3.5, read at $z = 1$; A06
α, β	random, global	SAG13 shape; posterior of an <code>emcee</code> chain, used only through $\rho_{\mathcal{B}}$	SAG13; A06
$\rho_{\mathcal{B}}$	deterministic	$\eta_{\mathcal{B}}/\eta_{\oplus}$, the SAG13 integral over $\mathcal{B}$ relative to the canonical box; independent of Γ	A06
$\eta_{\mathcal{B}}$	deterministic	$\eta_{\oplus} \rho_{\mathcal{B}}$	A06
$\mathbf{s}_i$	observed	$d, L_*, T_{\text{eff}}, R_*$, HIP id	HPIC (Tuchow+2024); A01
z_i	random, per star	exozodi level, $z_i \sim p_z$; four stars fixed at LBTI values; 200 joint draws of $\mathbf{z}$	Stark+2024 Sec. 3.3, Fig. 9; A02, A04
C_i	deterministic	planner’s completeness $C_i(\tau, k)$ at $A_G = 0.2$, over exposure and visits, counting only planets that are also characterizable	Stark+2019; A04
λ	deterministic	equal-slope multiplier closing the two-year budget	Stark+2014; A05
$\mathbf{a}_i$	decision	(τ_i, k_i) : the vertex of star i ’s concave cost envelope at slope λ	A05, A07
N_i	random, per star	$N_i \sim \text{Poisson}(\eta_{\mathcal{B}})$	Stark+2024 Sec. 3.1; A07
$\mathbf{o}_{ij}$	random, per planet	circular orbit: a, R from SAG13 over $\mathcal{B}$; $\cos i \sim \mathcal{U}$; independent epoch per visit	ExoVista/AYO; A04
A_{ij}	random, per planet	geometric albedo, $\mathcal{U}(0.08, 0.32)$, counted by Stark’s per-visit flux test	Stark+2024 Sec. 3.2; A02, A04
τ_{ij}	deterministic	detection time per visit (two channels in quadrature, S/N 7) and characterization time (R 140, best of nine bandpasses, best orbital phase)	Stark+2019 Eqs. 1–9; A04
D_{ij}	deterministic	$\mathbb{I}[\min_{v \leq k_i} \tau_{ijv}^{\text{det}} \leq \tau_i \wedge \tau_{ij}^{\text{char}} \leq 2 \text{ mo}]$	Stark+2019 Sec. 3; A04
Y	deterministic	$\sum_{i,j} D_{ij}$	A07

Table 1: Every node of Fig. 1, what it represents, and where it comes from. The labels A01–A15 are the pipeline steps (Table 6).

2.2 The joint distribution, and how it is marginalized

With the fixed quantities $(\mathcal{M}, \mathcal{B}, \{\mathbf{s}_i\})$ held constant, the graph factorizes as

$$p(\eta_{\oplus}, \alpha, \beta, \mathbf{z}, \mathbf{N}, \mathbf{o}, \mathbf{A}, Y) = p(\eta_{\oplus}) p(\alpha, \beta) \prod_i p_z(z_i) \text{Poisson}(N_i; \eta_{\mathcal{B}}) \times \prod_{j=1}^{N_i} p(\mathbf{o}_{ij} \mid \alpha, \beta, \mathcal{B}, \mathbf{s}_i) p_A(A_{ij}) \cdot \delta\left(Y - \sum_{ij} D_{ij}\right), \quad (1)$$

where the allocation enters D_{ij} through the deterministic function

$$\mathbf{a}_i = \mathbf{a}_i(\lambda(\eta_{\mathcal{B}}, \mathbf{C}), C_i), \quad C_i = C_i(\mathbf{s}_i, z_i, \mathcal{M}, \mathcal{B}, \alpha, \beta). \quad (2)$$

Two properties make this tractable.

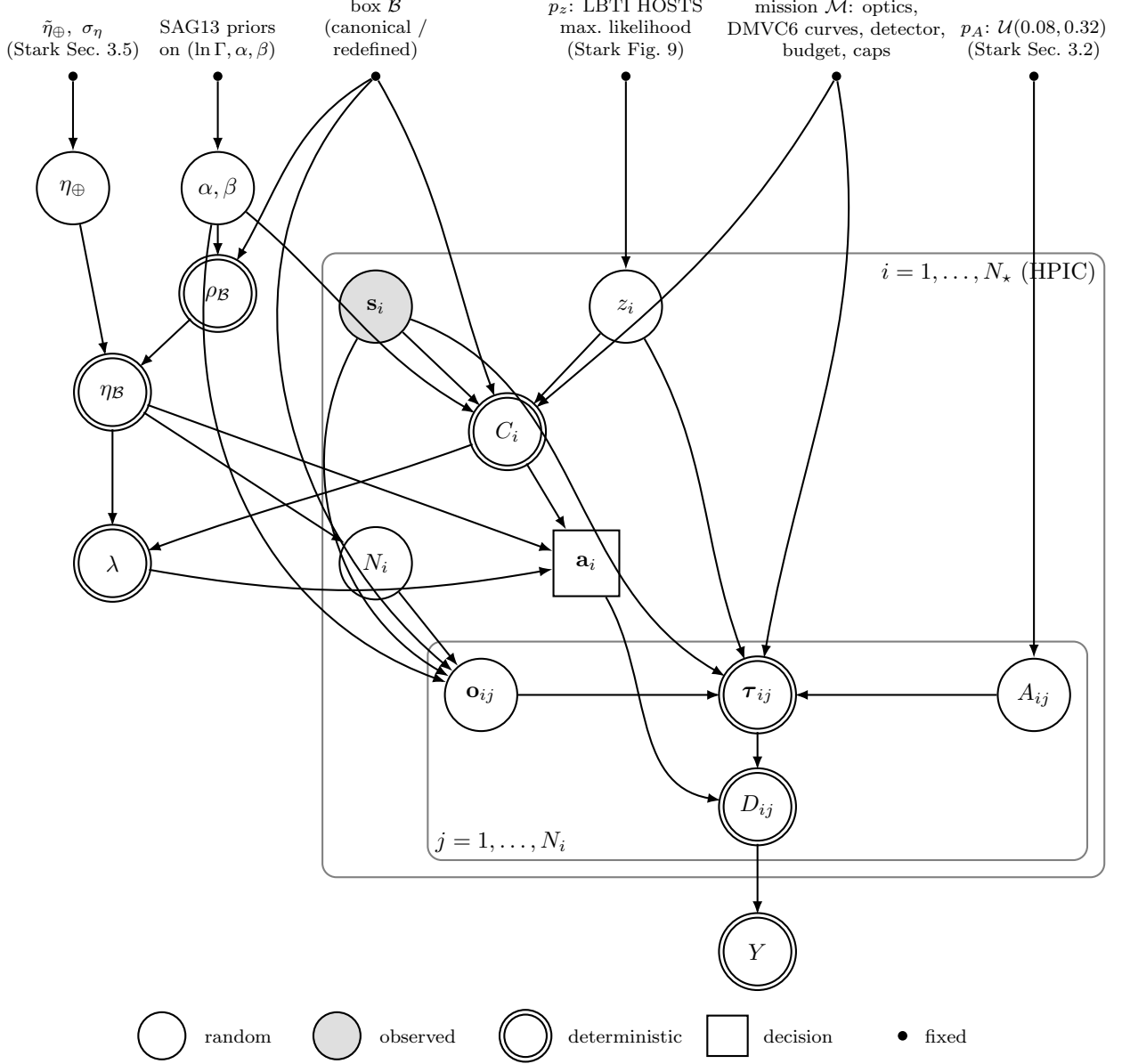


Figure 1: The yield model as an influence diagram. The outer plate runs over the $N_{\star}$ screened HPIC stars; the inner plate runs over the N_i planets that actually orbit star i . The only decision node is the survey allocation $\mathbf{a}_i = (\tau_i, k_i)$: the exposure time per visit and the number of visits. It is set by the equal-slope multiplier λ , which depends on the planner’s completeness curves C_i and on the occurrence rate in the box, $\eta_{\mathcal{B}}$. A planet is counted ($D_{ij} = 1$) if it is detected on some visit within its allocated exposure time *and* its spectrum can be taken within the two-month cap; the yield is $Y = \sum_{ij} D_{ij}$. Four stars whose exozodi has been measured by LBTI have z_i observed rather than drawn (not shown). The edges that control the uncertainty budget are those *entering* $\mathbf{a}_i$: $\eta_{\oplus}$ and z_i reach it (they are actionable), while A_{ij} and N_i do not (they are static).

Poisson thinning. Once the allocation is fixed, each planet is counted independently, with a probability that depends only on its host star, so the planet draw inside the plate can be marginalized exactly. If $N_i \sim \text{Poisson}(\eta_{\mathcal{B}})$ and each planet is counted with probability $\bar{C}_i$ (the completeness at the allocated exposure, averaged over albedo), then the number of counted planets around star i is $\text{Poisson}(\eta_{\mathcal{B}}\bar{C}_i)$, and the survey total is

$$Y \mid \eta_{\oplus}, \mathbf{z} \sim \text{Poisson}\left(\Lambda(\eta_{\oplus}, \mathbf{z})\right), \quad \Lambda = \eta_{\mathcal{B}} \sum_i \bar{C}_i(\mathbf{a}_i(\eta_{\mathcal{B}}, \mathbf{z})). \quad (3)$$

Step A07 evaluates Λ at 24 values of $\eta_{\oplus}$, *re-optimizing the survey at each one*, interpolates between them, and draws one Poisson count for each of the 14,400 posterior samples of $\eta_{\oplus}$. This is Stark’s construction: in his Sec. 3.1 he draws a Poisson number of planets for each star and counts those with a valid record of visits.

Albedo is static. The planner allocates time using C_i computed for a fixed albedo, $A_G = 0.2$. The yield is then counted from the same planets and orbits with each planet’s albedo A_{ij} drawn at random, using Stark’s per-visit test (Sec. 3.4). A planet darker than planned may go undetected in the fixed allocation. Because there is no path from A_{ij} into $\mathbf{a}_i$, the albedo penalty is a consequence of the graph, not an added correction.

What is marginalized.

- $\eta_{\oplus}$: marginalized, with the survey re-optimized for each value, because it is actionable.
- $N_i, \mathbf{o}_{ij}, A_{ij}$: marginalized exactly by thinning, Eq. (3).
- α, β : marginalized for the *ratio* $\rho_{\mathcal{B}}$; the planet population inside each box uses the SAG13 central values.
- $\mathbf{z}$: marginalized over 200 joint draws (Stark uses 500), with the survey *re-optimized for every draw*, because exozodi is actionable. Step A04 tabulates each star’s completeness at 23 exozodi levels (0–1024 zodis), and each draw interpolates between them; against direct evaluation at the drawn levels, this changes the optimized yield by at most 0.24%. Each posterior sample of $\eta_{\oplus}$ is paired with one draw of $\mathbf{z}$. At fixed $\eta_{\oplus}$, the yield averaged over draws is 17.73, with a draw-to-draw scatter of 0.57.

2.3 Actionable and static uncertainty

A source of uncertainty is *actionable* if there is a directed path from its node to the decision node, and *static* if there is not. In Fig. 1:

- $\eta_{\oplus} \rightarrow \eta_{\mathcal{B}} \rightarrow \lambda \rightarrow \mathbf{a}_i$: actionable. A mission that finds more planets must spend more time characterizing them, so the yield-maximizing survey depends on $\eta_{\oplus}$. This is why yield grows more slowly than $\eta_{\oplus}$, and why the redefined yield cannot be obtained by rescaling.
- $z_i \rightarrow C_i \rightarrow \mathbf{a}_i$: actionable. Exozodi is measured on the first visit, and the survey adapts to it (Stark et al. 2015).
- $A_{ij}, N_i, \mathbf{o}_{ij}$: static. The planner cannot know which stars host planets, where those planets are, or how dark they are.

Each observational penalty in this problem (the albedo penalty, the exozodi penalty, the difference between Stark’s red and purple curves in his Fig. 10) is a statement about which of these paths exist.

3 Model components

3.1 Stars

The target list is the HWO Preliminary Input Catalog (HPIC; Tuchow et al. 2024), which Stark et al. (2024) use. It contains 11,757 dwarf stars within 50 pc, 7,767 of them FGK. Each star’s photon flux is computed from a blackbody at its catalogued T_{eff} and R_* . Compared with the measured V magnitudes of 674 stars in HPIC, the median residual between 5300 and 7300 K is +0.019 mag, and the fluxes agree with the Pickles spectral templates used by EXOSIMS to a ratio of 0.979. The survey chooses from the 1,500 stars whose habitable zones are widest in units of λ/D , after a noise-floor screen that depends on the box being searched.

3.2 Occurrence: $\eta_{\oplus}$

Stark et al. (2024, Sec. 3.5) derive $\eta_{\oplus}$ by integrating the Kepler DR25 posteriors of Bryson et al. (2021) over their canonical domain, drawing equally from Bryson’s two bounding cases for extrapolating completeness, and quote $0.26^{+0.29}_{-0.14}$ as “mean and 86% confidence interval”. Bryson’s posterior chains are not published, so the distribution must be reconstructed.

We use a lognormal and set its width from Stark’s own Fig. 10, which shows the yield distribution with $\eta_{\oplus}$ fixed (red curve, mean 17.35) and with $\eta_{\oplus}$ uncertain (purple curve). Holding this model’s yield-versus- $\eta_{\oplus}$ relation at the level of the red curve, the lognormal that best turns the red curve into the purple one has median 0.26 and ln-width 0.80, reproducing the purple curve to a total-variation distance of 0.019. Reading the quoted interval as 86% gives a ln-width of 0.52, a distance of 0.142, and a peak at 15 instead of the published 10. Reading it as a one-sigma (68%) interval gives a ln-width of 0.76, a distance of 0.024, and a peak at 12. Bryson’s Fig. 14, digitized from its vector source, shows each extrapolation case with a ln-width of about 0.8, consistent with Stark’s figure and not with the 86% reading. We therefore read the interval at $z = 1$ (Table 1).

This choice departs from Stark’s text in two places: the interval is read as 68% rather than 86%, and the lognormal is centered on 0.26, which makes 0.26 its median, whereas Stark calls 0.26 the mean (the model’s mean $\eta_{\oplus}$ is 0.343). Step A15 tests the interval against its own source without using the yield model. It mixes Bryson’s two cases equally, as Stark describes, and rescales the mixture so that its mean is 0.26. That distribution puts 59% of its probability inside the quoted [0.12, 0.55], much closer to 68% than to 86%; its own 86% interval, [0.047, 0.653], is far wider than the one printed. Its median, 0.183, lies well below its mean, so no lognormal can match the stated mean, interval and confidence level at once. The 68% reading may therefore correct a transposed digit, or it may supply a width that stands in for a tail the lognormal lacks; the figure cannot distinguish the two. Using the rescaled Bryson mixture itself as the $\eta_{\oplus}$ law matches Fig. 10 poorly (Sec. 6), probably because Bryson’s published distribution is for a different selection of planets and stars than the one Stark integrated over. We keep the 68% lognormal and treat the reading as an open question.

The redefined box enters only through ρ_B , the ratio of the SAG13 integrals over the two boxes, which does not depend on the normalization Γ . Its posterior comes from an **emcee** chain over $(\ln \Gamma, \alpha, \beta)$ with SAG13 priors, and it is narrow because Γ cancels in the ratio.

3.3 Exozodiacal dust

Each star’s exozodi level is drawn from the LBTI HOSTS maximum-likelihood distribution (Ertel et al. 2020) that Stark et al. (2024) use, digitized from the vector data of their Fig. 9 (red curve). One zodi is the dust level of the Solar System’s own zodiacal cloud. The distribution has a median of 2.98 zodis (Stark states three), and 21% of draws lie above 100 zodis. The digitized histogram accounts for 0.948 of the $10^4 \times 500$ draws it implies; we renormalize to make up the shortfall, which

reproduces the stated median. As Stark does, whenever levels are drawn we pin four stars to their LBTI measurements: ϵ Eri (297 zodis), θ Boo (148), 72 Her (588; listed as “w Her” in HPIC) and 110 Her (235).

The heavy tail matters more than the median, because a star with hundreds of zodis is effectively removed from the survey. The exozodi surface brightness follows Stark (2014, App. C): a constant optical depth in the habitable zone, evaluated at the Earth-equivalent insolation distance (EEID), with solar colors.

3.4 Albedo

Planets have geometric albedos drawn uniformly between 0.08 and 0.32 (Stark et al. 2024, Sec. 3.2). Following Stark, the observation plan is fixed at $A_G = 0.2$, and a planet with a drawn albedo is counted only if the fixed plan would have detected it: its albedo-adjusted flux must lie inside the range of fluxes the visit actually reached. That range is bounded below by the faintest flux the exposure detected and above by the brightest flux the inner working angle allowed (“the faintest detected planet flux along the orbit segment ... is limited by the exposure time, while the brightest planet flux along the segment ... is constrained by the IWA of the coronagraph”). Both bounds are applied. Characterization is budgeted at $A_G = 0.2$, as Stark (2024, Sec. 3.2) states.

With this test the albedo penalty on the expected yield is 3.6%, against the 12% that Stark reports. The alternative of re-deriving each drawn-albedo planet’s exposure time gives 8.6%, so neither method reproduces the published value here (Sec. 6).

3.5 Radiometry and exposure times

Exposure times follow Eq. 1 of Stark et al. (2019),

$$\tau = (S/N)^2 \frac{CR_p + 2 CR_b}{CR_p^2}, \quad (4)$$

where S/N is the required signal-to-noise ratio, CR_p is the planet’s count rate, and CR_b is the background count rate. The background includes leaked starlight, local zodiacal light, exozodiacal light, and detector noise (dark current, and clock-induced charge with the Geiger factor of 6.73 from Stark 2019, Eq. 9). Local zodiacal light follows Stark et al. (2014): the Leinert et al. (1998) spectrum, scaled by 0.69 to a solar elongation of 135° and adjusted for each star’s ecliptic latitude. This matches the zodi in AYO’s exposure-time benchmark (Stark et al. 2025) to 0.93–1.01.

Both sky backgrounds are multiplied by the extended-source throughput T_{sky} of Stark (2019, Eqs. 5–6), “the instrument’s throughput for extended sources”, computed “by first convolving the spatially-dependent PSF at all locations with a normalized uniform background”. This is a function of position: background light near the inner working angle (IWA) is attenuated by the same focal-plane mask that attenuates a planet there. Stark’s map is not published, so we reconstruct it as $T_{\text{sky}}(r) = 0.678 \Upsilon_c(r)/\Upsilon_{c,\text{max}}$, the core-throughput curve rescaled to its large-separation value $0.678 = \Upsilon_{c,\text{max}}/\text{EE}(0.7 \lambda/D)$. This assumes that the fraction of an off-axis PSF falling inside the photometric core does not change with separation, and it omits the $\sim \lambda/D$ smoothing of the convolution.

The coronagraph is the DMVC6 of Stark et al. (2024), digitized from the vector data of their coronagraph figure. Its curves are evaluated in *circumscribed* λ/D , the convention the figure uses, and the collecting area is the full obscured primary mirror. Detection combines the 450 nm and 550 nm channels in quadrature and requires S/N 7. The noise floor at $\Delta\text{mag} = 26.5$ enters as a systematic term in the exposure-time equation, which lengthens exposures as a planet approaches it, rather than as a hard cut. The two-month limits on detection and characterization include overheads (Stark et al. 2024, Table 2). Characterization is an $R = 140$ spectrum from the integral field spectrograph (IFS), at the S/N per spectral bin given in Stark’s Fig. 3, using whichever of nine

bandpasses is best, with 96 pixels per spectral bin at $1\,\mu\text{m}$, taken at the planet’s most favorable orbital phase.

No calibration. The model contains no throughput factor fitted to Stark’s results. His 22.5 candidates are an output of AYO, not an input, and a scalar fitted to them would absorb any error that changes exposure times uniformly: at a refitted factor, every alternative treatment we tested changes per-planet exposure times by a nearly uniform 1.0 ± 0.05 , and the yield’s weak dependence on throughput (an elasticity of about 0.4) lets a fitted factor cancel it entirely. For reference, the factor that would reproduce 22.5 is 1.20; step A03 records it as a diagnostic only.

3.6 Completeness and the survey allocation

For each star, 1 200–3 000 planets are injected across the box and propagated through the radiometry at up to six independent visit times. The completeness $C_i(\tau, k)$ is the fraction of injected planets that are detected within exposure τ on at least one of k visits *and* can be characterized within two months; Stark et al. (2019) state that planets failing either limit “did not count toward the yield”. The survey then maximizes $\sum_i \eta_{\mathcal{B}} C_i$ within a two-year budget. Each star is charged $k(\tau'\tau + t_{\text{oh}})$ for detection, plus $\eta_{\mathcal{B}} C_i (\tau'\bar{\tau}_{\text{char}} + t_{\text{oh}})$ for characterizing whatever it finds, where τ' is the time multiplier for wavefront control and t_{oh} is the slew and wavefront-control overhead per observation. Pooling every (k, τ) option into a single concave envelope and sweeping the equal-slope multiplier λ reproduces an exact dynamic-programming optimum to 0.00%.

4 Validation against the published record

Step A09 compares every quantity it can against the published record and records each check’s source and tolerance. Checks against quantities that entered the model as inputs (the round trips through the digitized coronagraph curves) are labeled *tuned* and excluded, leaving the 24 independent checks of Table 2, of which 21 agree. The three that disagree are the albedo penalty (Sec. 3.4) and the characterization times at 6 and 9 m (Sec. 7).

The yield level. The expected 6 m yield with exozodi and albedo fixed is 20.92 against the published 22.5 (−7.0%). With $\eta_{\oplus}$ fixed and all other uncertainties included, the realized yield has mean 17.73 against the 17.35 of Stark’s red curve (2.2%), and the exozodi penalty is 12.2% against 11.1%.

The realized-yield distribution. The purple curve in Stark’s Fig. 10, which includes every astrophysical uncertainty, peaks at 10, with median 18 and mean 21.0. Our digitization reproduces three numbers quoted in his text (the red-curve mean of 17.3, and P_{25} values of 6% and $\sim 32\%$), so the published curves are known to within the width of a plotted line. This model’s realized distribution has mean 22.4, median 19, and 0.26 of its probability below 12 candidates (published 0.29); its total-variation distance from the purple curve is 0.037. Its peak is at 12, computed from the exact Poisson mixture over the sampled expected yields; the mode of a histogram of realized draws (11) is too noisy for this comparison, since bootstrapped it cannot distinguish 10 from 16. Scaling the overall yield level by 5–10% shifts the exact peak by one bin, which sets the resolution of this statistic. Figure 10 shows the two distributions side by side.

The aperture scaling. With nothing fitted at any aperture, the model gives P_{25} (including the spread in $\eta_{\oplus}$) of 35 / 53 / 66 / 76% at 6, 7, 8 and 9 m, against a published 32 / 53 / 67 / 78%. The expected yield at $\eta = 0.24$ runs 17.8 / 24.4 / 30.6 / 37.1, scaling as $D^{1.81}$ against the published $\sim D^{1.9}$.

An independent implementation. EXOSIMS (Savransky & Garrett 2016) shares no code or scheduling method with this pipeline. Given the same FGK stars and optics, its count rates agree with ours to ratios of 1.001 (local zodi), 0.975 (exozodi) and 0.979 (star). It cannot be run end to end in this environment, because of version incompatibilities that also break its own bundled

Check	Published	Model	Diff.	Tol.	Verdict
eta.Earth over the canonical box (SAG13)	0.24	0.2404	0%	3%	agrees
Uncalibrated 6 m yield	22.5	20.92	7%	20%	agrees
Throughput factor needed to reach 22.5	1	1.195	20%	100%	agrees
Sampling distribution mean (Fig. 4)	22.5	20.91	7%	20%	agrees
Sampling distribution sigma (Fig. 4)	5	4.567	9%	25%	agrees
Albedo penalty on expected yield	0.12	0.03608	70%	50%	disagrees
P25 including sigma_eta, 6 m	0.32	0.3514	10%	25%	agrees
Yield-aperture exponent	1.9	1.813	5%	15%	agrees
Exozodi-sampling penalty on expected yield	0.111	0.1217	10%	50%	agrees
P25 excluding sigma_eta, 6 m	0.06	0.06264	4%	50%	agrees
Mean characterization time, first 18 EECs, 6 m (days)	22	14.34	35%	30%	disagrees
Mean characterization time, first 18 EECs, 9 m (days)	3.5	2.053	41%	30%	disagrees
Characterization-time ratio 6 m / 9 m	6.2	6.987	13%	20%	agrees
P25 including sigma_eta at 6 m	0.32	0.3508	10%	25%	agrees
P25 including sigma_eta at 7 m	0.53	0.5271	1%	25%	agrees
P25 including sigma_eta at 8 m	0.67	0.6573	2%	25%	agrees
P25 including sigma_eta at 9 m	0.78	0.755	3%	25%	agrees
Fixed-eta yield mean (Fig. 10 red)	17.35	17.73	2%	10%	agrees
Realized-yield distribution median	18	19	6%	15%	agrees
Realized-yield fraction below 12 EECs	0.2907	0.2597	11%	35%	agrees
Realized-yield distribution mean	21	22.45	7%	30%	agrees
Realized-yield distribution mode	10	11	10%	25%	agrees
Maximum luminosity among reachable targets	20	12.5	38%	60%	agrees
Maximum distance among reachable targets	25	22.02	12%	60%	agrees

Table 2: The 24 independent checks of step A09, of which 21 agree. Published values from Fig. 10 are digitized from the figure’s vector data: the fixed- η mean comes from Stark’s red curve, and the median, mode and low tail from his purple curve.

example script.

The published figures. Nine of the 26 figures in Stark et al. (2024) show a quantity this model computes under the same assumptions. Appendix A reproduces each beside the original, drawn with the published axes, ranges and line styles and with nothing rescaled.

5 Results

Table 3 gives the yields for both boxes.

The yield ratio is the most robust result. It is computed at a single aperture within a single implementation, so an error in the absolute throughput appears in both numerator and denominator, and it stays between 0.2201 and 0.2386 across the model variants of Sec. 6. The absolute yields are less secure, because they depend on the characterization-time discrepancy of Sec. 7.

5.1 Why the yield ratio exceeds the occurrence ratio

The occurrence ratio, 0.204, splits into two factors: about 0.32 from narrowing the habitable zone and about 0.63 from selecting by mass. The yield ratio, 0.232, is larger, for three reasons, each visible as an edge in Fig. 1:

1. **Characterization time is freed** ($\eta_B \rightarrow \lambda$). Fewer expected detections leave more of the budget for searching. This raises the ratio.
2. **The faintest planets are removed** ($\mathcal{B} \rightarrow \mathbf{o}_{ij}$). Cutting the zone off at 1.20 scaled AU removes planets up to $(1.67/1.20)^2 \approx 1.9$ times fainter than those that remain. This also raises the ratio.

	Canonical box	Redefined box
Habitable zone (AU, scaled by $\sqrt{L_\star}$)	0.95–1.67	0.96–1.20
Planet selection	$0.8 (a/\text{EEID})^{-1/2} < R/R_\oplus \leq 1.4$	$0.5 \leq M/M_\oplus \leq 2, R = M^{0.27}$
Occurrence η (median, 68%)	$0.255^{+0.282}_{-0.134}$	$0.0520^{+0.0577}_{-0.0274}$
Realized yield, median	18.8	4.3
Realized yield, mean	22.5	5.6
Realized yield, 90% interval	5.6–53.4	1.2–14.8
$P(\geq 25 \text{ candidates})$	35.1%	1.20%
Occurrence ratio, redefined/canonical	$0.204^{+0.017}_{-0.016}$	
Yield ratio, redefined/canonical	$0.232^{+0.027}_{-0.021}$	

Table 3: Yields for the baseline design, a LUVOIR-B-like telescope 6 m in inscribed diameter. Yields are realized counts of candidates, marginalized over the uncertainty in $\eta_\oplus$, the random draw of planets, their geometric albedo, exozodi, and the survey’s re-optimization. This is the quantity Stark et al. (2024) plot in their Fig. 10, and from which they compute P_{25} , the probability of finding at least 25 candidates. The occurrence rates and both ratios are medians with 68% intervals.

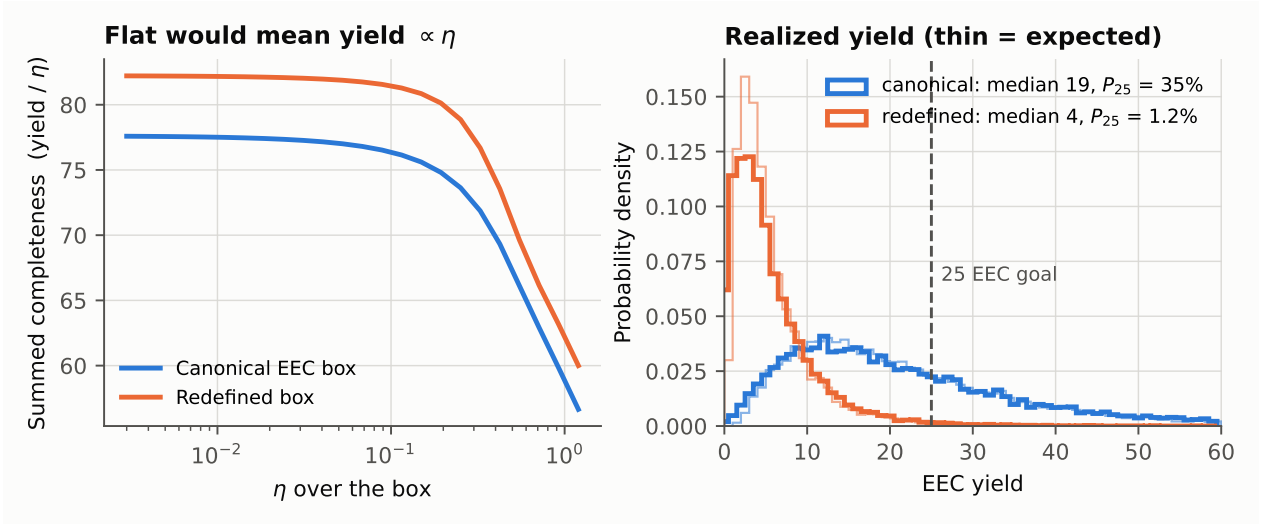


Figure 2: Left: summed completeness as a function of η . It declines because characterizing more planets eats into the time available for searching; this is the $\eta_\oplus \rightarrow \lambda$ edge of Fig. 1. Right: the realized yield distributions from Eq. (3), marginalized over $\eta_\oplus$ (heavy lines), and the expected yield Λ alone (thin lines).

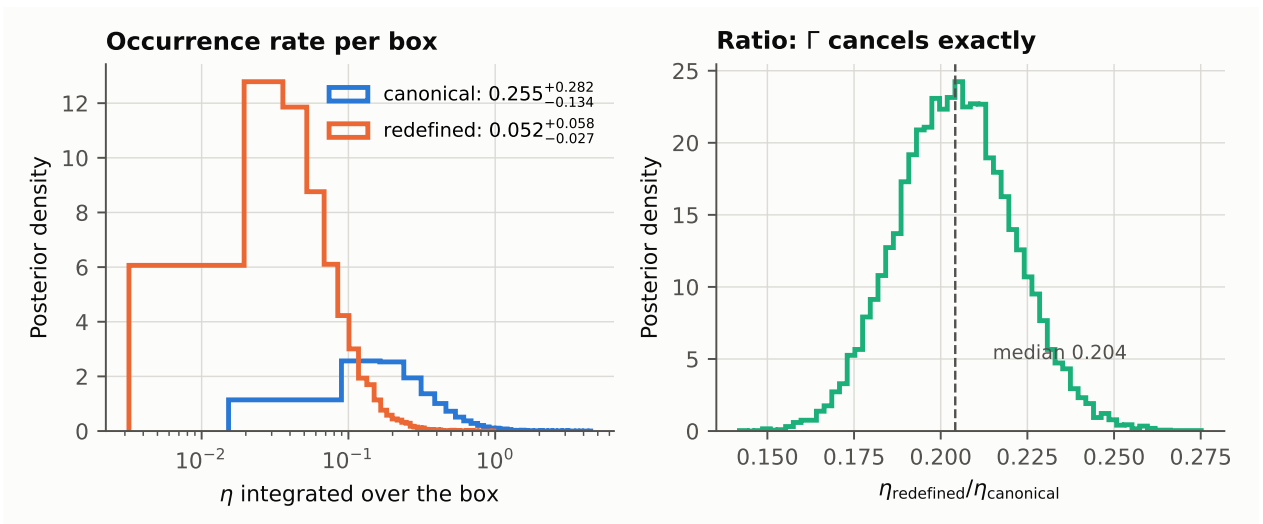


Figure 3: Left: the occurrence rate integrated over each box. Right: their ratio ρ_B , which is tightly constrained because Γ cancels.

3. **Distant stars are lost** ($\mathcal{B} \rightarrow C_i$). Around distant stars, only the outer part of the habitable zone lies outside the inner working angle, and that is the part the redefined box drops. This lowers the ratio.

The first two effects win. Scaling the canonical yield by the occurrence ratio alone would be wrong by 13%. Even so, under the redefined selection a 6 m telescope reaches 25 candidates with a probability of only 1.20%: with a narrow habitable zone and a mass-selected sample, the decadal survey’s benchmark is out of reach at this aperture for any plausible value of $\eta_{\oplus}$.

5.2 The Earth zone and the classic habitable zone, star by star

In this section, the *Earth zone* is the redefined box and the *classic habitable zone* is the canonical box. Step A11 tabulates both zones for each of the 1,500 screened stars, in AU, in milliarcseconds, and in λ/D for the 6 m telescope at 550 and 1000 nm, together with how many candidates each zone’s own optimized survey expects to find there. Each zone’s survey is re-optimized in every exozodi draw at that zone’s posterior-median occurrence rate ($\eta = 0.255$ and 0.0520), and a star’s expected yield is η times its completeness averaged over the draws. The full table is `TabulateEarthZones/earthZoneTable.csv`; Appendix C lists the 172 stars that contribute to either survey.

Totals. Measured in scaled AU, the Earth zone is a third as wide as the classic zone, and it holds 0.204 as many candidates per star. Summed over all stars, the expected yield is 18.76 in the classic zone and 4.26 in the Earth zone, a ratio of 0.227. Taken star by star, the ratio has a median of 0.228, with 68% of stars between 0.169 and 0.305. The classic survey observes 180 stars in at least half of the draws, reaching a mean completeness of 0.42; the Earth-zone survey observes 157 stars at 0.53. The narrower zone is easier to complete, because its planets orbit closer in and are therefore brighter, so the survey concentrates on slightly fewer stars and finds a larger fraction of their candidates. Table 4 shows how the ratio varies with luminosity and distance: around distant and luminous stars the outer planets of the classic zone are too faint to find anyway, so losing them costs little.

Angle. Because the Earth zone sits at the inner edge of the classic zone, it is the part of the classic zone that is harder to reach in angle. Taking the inner working angle as the separation where throughput falls to half, $2.93 \lambda/D$ (inscribed), on average 91% of the Earth zone lies outside it at 550 nm, compared with 96% of the classic zone. At 1000 nm, where water is sought, the figures are 47% and 63%. Both are averaged over the stars that either survey observes. The coronagraph still passes some light inside the inner working angle, which is why the loss in yield is smaller than the loss in angle.

6 Sensitivity to settings the published record leaves open

Three settings are not fully determined by the published description: the reading of the $\eta_{\oplus}$ interval (Sec. 3.2), the albedo method (Sec. 3.4), and the reconstruction of T_{sky} (Sec. 3.5). The adopted model follows Stark’s documented method for the albedo and uses the sourced T_{sky} term; it departs from his text only in the interval reading. Step A14 re-runs the pipeline end to end for six variants, each differing from the adopted model in named settings, with every other setting, input file and random seed held fixed (Table 5, Fig. 5).

The main result does not depend on these settings. Across all seven models the yield ratio lies between 0.2201 and 0.2386, inside the 68% interval of the adopted value.

Agreement with Fig. 10 depends on one setting. The adopted model and the three variants that change the albedo method, the interval reading, or both form a complete 2×2 . Decomposing the distance to Fig. 10 over that square, the interval reading contributes +0.0936, the albedo method

Luminosity ($L_{\odot}$)	stars	stars observed		expected EECs		ratio
		classic	Earth	classic	Earth	
$L < 0.3$	47	36	33	5.76	1.13	0.20
0.3-0.6	75	31	25	4.34	0.92	0.21
0.6-1	102	19	15	1.96	0.42	0.21
1-2	294	33	28	3.15	0.75	0.24
2-5	665	44	40	3.05	0.89	0.29
$L > 5$	317	17	16	0.50	0.16	0.32
Distance						
< 5 pc	15	15	15	3.32	0.71	0.21
5-10 pc	59	59	57	9.69	2.12	0.22
10-15 pc	95	66	54	4.69	1.20	0.26
15-20 pc	133	40	31	1.05	0.23	0.22
> 20 pc	1198	0	0	0.02	0.00	0.03

Table 4: Where each zone’s yield comes from, by stellar luminosity and by distance. A star counts as observed if it is allocated time in at least half of the exozodi draws. Each zone’s survey is optimized for its own candidates.

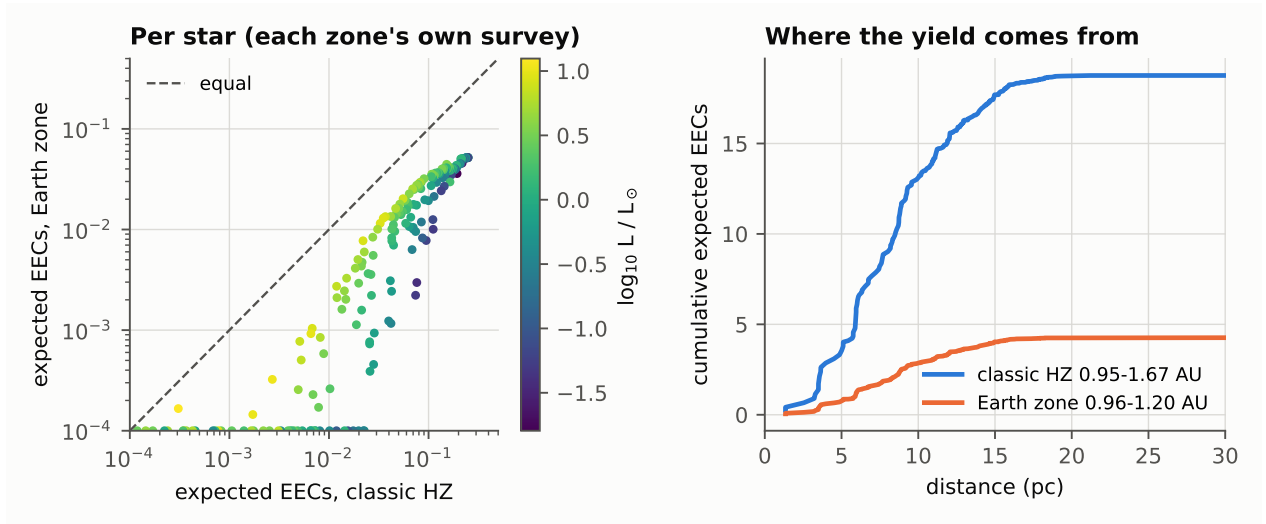


Figure 4: Left: each star’s expected Earth-zone yield plotted against its classic-zone yield, colored by luminosity. Right: cumulative expected yield as a function of distance, for both zones.

Model	Peak	Mean	$P(Y < 12)$	Albedo pen.	Distance	Yield ratio
Adopted	12	22.4	0.26	3.6%	0.037	0.2315
$\eta_{\oplus}$ interval read as 86%	15	20.6	0.19	3.6%	0.131	0.2292
Albedo: exposure re-derived	11	21.0	0.28	8.6%	0.038	0.2386
Albedo re-derived and 86% interval	15	19.5	0.22	8.6%	0.132	0.2363
Constant T_{sky}	10	19.2	0.34	2.4%	0.089	0.2225
86% interval and constant T_{sky}	12	17.4	0.30	2.4%	0.146	0.2201
Rescaled Bryson mixture for $\eta_{\oplus}$	7	17.8	0.41	3.6%	0.155	0.2279
Stark et al. (2024)	10	21.0	0.29	12%	—	—

Table 5: The adopted model and the six variants of step A14. “Peak” is the mode of the exact Poisson mixture; the distance is the total-variation distance between the model’s realized-yield distribution and Stark’s Fig. 10 purple curve (smaller is better). The last column is this report’s main result.

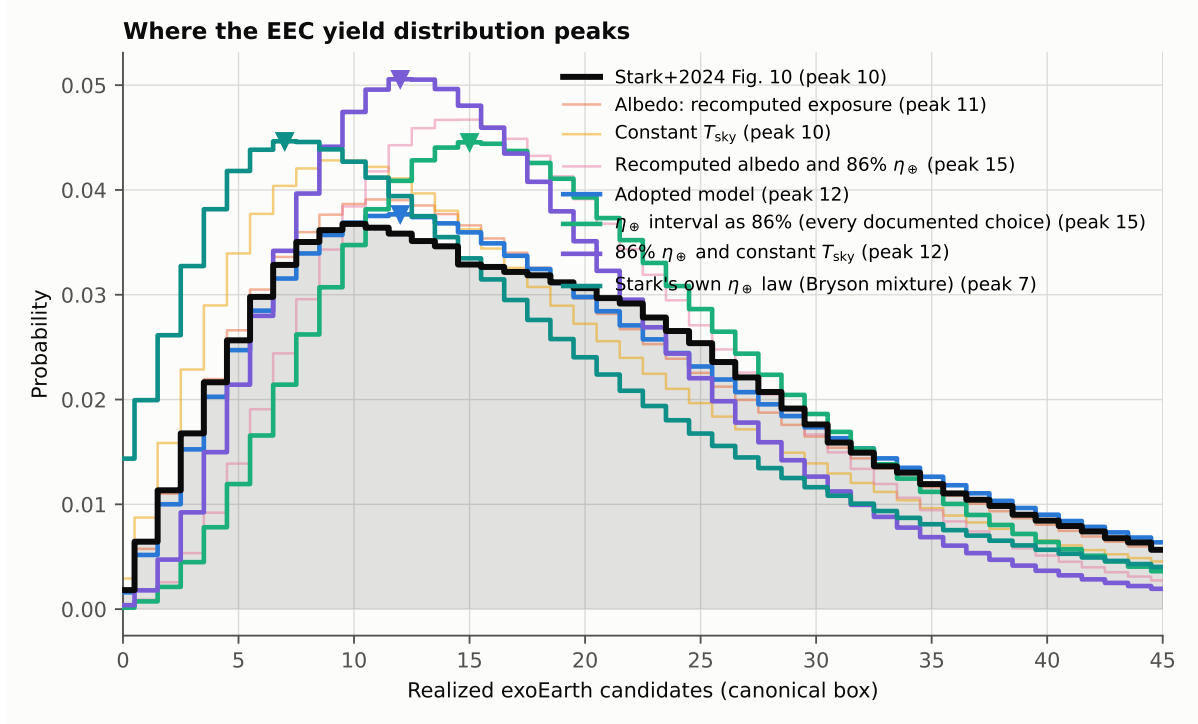


Figure 5: Realized yield distributions for every model of Table 5, against the digitized Stark et al. (2024) Fig. 10 (heavy black). Each curve is the exact Poisson mixture over that model’s sampled expected yields. Single-setting variants are drawn faintly; the others carry peak markers.

-0.0009, and their interaction +0.0006: the interval reading accounts for 99.1% of the disagreement and the albedo method 0.9%.

Following every documented choice worsens the agreement. With the albedo test already Stark’s, reading the interval as the stated 86% gives a model that follows every documented choice. It moves the peak from 12 to 15 (published 10) and the distance from 0.037 to 0.131. Either this model is wrong in a way that only the 68% reading offsets, or the published text and figure are not mutually consistent and the figure is the better guide to what was computed. The test of Sec. 3.2 favors the second, but nothing available rules out the first. The rescaled Bryson mixture, which removes both the 68/86 choice and the mean/median ambiguity, does not resolve it: the three candidate laws bracket Stark’s curve, with peaks at 7, 12 and 15 against 10, and the published information cannot choose among them.

The albedo penalty. The two albedo methods match Fig. 10 about equally well (distances 0.037 and 0.038), but they differ in the penalty: 3.6% with Stark’s per-visit test and 8.6% with re-derived exposures, against his reported 12%. Stark’s described method does not reproduce his reported penalty in this model.

7 The characterization-time discrepancy

The yield level, the exozodi penalty, the aperture scaling of P_{25} and the shape of the realized-yield distribution all agree with Stark’s (Sec. 4). One discrepancy remains, and it appears in two ways.

Characterization times. The mean characterization time of the first 18 EECs is 14.3 days at 6 m and 2.05 days at 9 m, against Stark’s 22 and 3.5 (his Sec. 4.1); the ratio between the two apertures is 7.0 against his 6.2. The distributions show where the difference lies (Fig. 13). At every aperture, Stark’s first 18 EECs include a plateau of spectra taking 15–50 days, whereas in this model the individual characterization times fall away steeply, with almost none above about 15 days at 9 m.

The target mix. Compared with Stark’s Fig. 11 (our Fig. 11), using the same characterization-gated completeness and one representative exozodi draw, this model agrees for the nearest stars: within 6 pc the median completeness is 0.98 against a published 0.95. Further out the two diverge: stars at 10–15 pc have a median of 0.35 (published 0.73), and stars at 15–20 pc 0.07 (0.28). By luminosity, stars of 1–2 $L_{\odot}$ have 0.08 (published 0.62), and stars of 2–5 $L_{\odot}$ have 0.14 (0.47), with 42% of the latter allocated essentially no time. This model’s survey concentrates on nearby stars instead, and reaches Stark’s total yield by completing them more thoroughly.

The mechanism. At $R = 140$ near $1 \mu\text{m}$, the background during characterization is dominated by exozodi. The exozodi count rate is set by a surface brightness that is nearly the same for every star, while the planet dims as d^{-2} , so characterization time grows roughly as d^4 . For an Earth twin at quadrature, the ratio of exozodi counts to planet counts per spectral bin is 1.3 at τ Cet, 7 at 61 Vir, 20 at θ Boo and 36 at 110 Her, and the corresponding characterization times are 0.21, 10, 72 and 255 days, against a 60-day cap. Distant, luminous targets are priced out of the survey, while nearby ones are characterized in hours to days.

AYO’s own exposure-time calculator agrees with these times. The Stark et al. (2025) benchmark gives Earth-twin water spectra of 37 days at 12.3 pc and 104 days at 18.2 pc, for a telescope slightly better than the baseline, and given that benchmark’s inputs our code reproduces its per-observation times to 0.94–1.02. The exposure-time equations are therefore not where the two models differ.

Explanations ruled out by direct test.

- *The noise floor.* Removing it entirely leaves the completeness of luminous stars unchanged.
- *The detector saturation definition.* Stark (2019) sets the saturation rate CR_{sat} to ten times an Earth twin’s count rate in its core pixel, whereas this model uses the whole scene. Switching definitions changes characterization times by less than 1%.
- *The number of IFS pixels.* Our 96 per spectral bin matches Stark (2024, Table 2).
- *Detector noise in general.* Stark’s Scenario D (a Skipper CCD) shortens characterization times by a factor of 1.5, which his text attributes mostly to a 30% gain in throughput. The share left for the detector, about 12%, matches this model’s.
- *Which targets count as the first 18.* Ordering targets by survey priority rather than by characterization cost changes the mean by 1%.
- *The metric used for Fig. 11.* Detection-only completeness fixes the low-luminosity stars but not the luminous ones.
- *Dust color.* The zero point is a solar-colored source normalized in the V band, the treatment Stark (2015) describes.
- *Exozodi brightness at the planet’s own separation.* Stark (2019, Table 1) notes that it “varies with spectral type and planet-star separation”; evaluating it there rather than at the EEID leaves the first-18 times essentially unchanged.
- *Charging characterization per star.* Budgeting spectra per star, as in Stark et al. (2015), moves the yield with $\eta_{\oplus}$ fixed the wrong way and leaves the target mix unchanged.
- *The orbital phase of the spectrum.* Taking spectra at the detection phase rather than the most favorable one lengthens the times only part of the way.

No throughput setting supplies the difference. Four parameters act on throughput: a factor on all bands (κ), a separate factor on the spectral bands only (κ_c), the normalization of the coronagraph leak term, and the radial shape of T_{sky} . Step A13 evaluates a 56-point design over that four-parameter box, recording at every point the 6 m planning yield, the characterization times at 6 and 9 m, and the Fig. 11 completeness by distance (Fig. 6).

- Across all 56 points, the first-18 characterization time at 6 m never exceeds 16.9 days, against the published 22.
- Of the 20 points that reproduce the published 6 m yield to within 10%, the first-18 time spans only 7.0–11.6 days at 6 m and 0.79–1.49 days at 9 m, overlapping neither published value.
- Constrained on the yield alone, the fit gives $\kappa = 1.44$ and $\kappa_c = 1.07$, with a correlation of -0.79 between their logarithms, and forces the 6 m characterization time to 9.7 days (6.4–12.7).
- The singular values of the local sensitivity are 31.72, 2.31, 1.50, 0.90. The first is 13.7 times the second, so these observables constrain essentially one combination of the four parameters: the anticorrelation between κ and κ_c .

Characterization time can be bought within this family only by giving up yield: when every published observable is imposed at once, the fit pushes κ_c onto the lower edge of its allowed range and the yield falls to about 16.

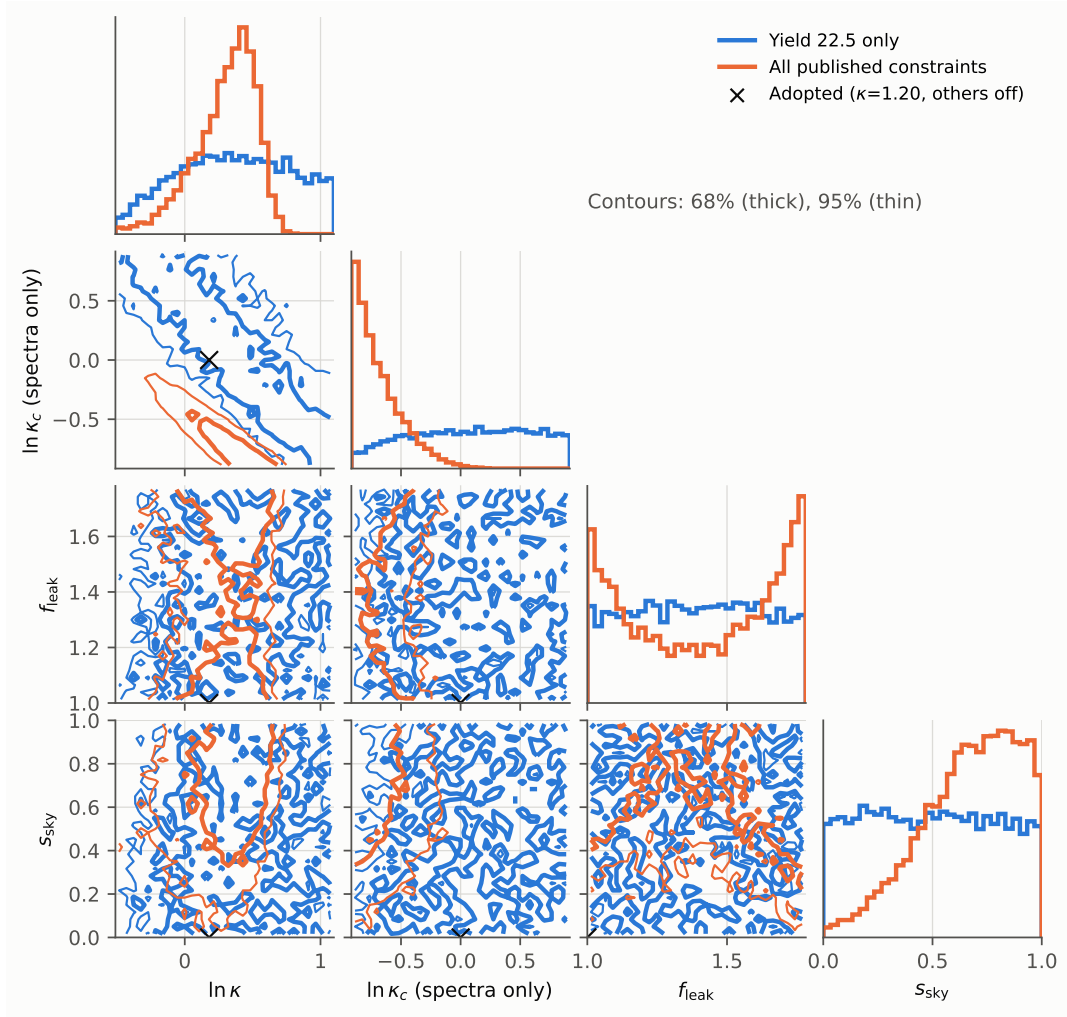


Figure 6: Posteriors over the four throughput parameters, from a quadratic emulator of the 56-point design of step A13; the emulator’s leave-one-out error is added to the tolerance, so emulator error cannot pose as a constraint. Blue: constrained on the published 6 m yield alone. Orange: constrained on the yield, the characterization times at 6 and 9 m, and the Fig. 11 completeness in three distance bins. The $\ln \kappa$ – $\ln \kappa_c$ panel shows the degeneracy the published numbers leave open; the orange contours run into the lower bound on κ_c while still failing to reach the published characterization times.

8 Discussion

8.1 Two questions for the AYO authors

Both remaining disagreements reduce to questions of fact that the published description does not settle.

Question 1: how does the AYO baseline survey characterize its 10–20 pc targets within the two-month limit? Stark’s Fig. 11 shows habitable-zone completeness of 0.5–0.8 for these targets, while AYO’s own calculator gives their Earth-twin spectra 37–104 days, and no setting of the throughput parameters reconciles the two (Sec. 7). Either those stars are characterized under conditions more favorable than an Earth twin at maximum elongation, or the completeness in Fig. 11 is not gated on characterization, or characterization is charged to the budget differently from the way this model charges it. Whether the S/N per spectral bin refers to a single bin or to a band average is a further detail the papers leave open.

Question 2: is $\eta_{\oplus} = 0.26$ the mean or the median, and is $[0.12, 0.55]$ an 86% or a 68% interval? The text says mean and 86%. The median-and-68% reading reproduces Fig. 10, and the stated reading does not (Sec. 6); the quoted interval holds 59% of its own source distribution.

These questions bear directly on mission design. The characterization budget sets the target mix, and the target mix sets which stars need exozodi measurements in advance. The $\eta_{\oplus}$ interval sets the width of the yield distribution, and the width sets $P(Y \geq 25)$, the quantity the decadal recommendation is written in. Neither discrepancy is known to be an error in AYO; each is equally consistent with an error in this reimplementation that has not been found. What the evidence supports is narrower: the published description does not determine these two quantities.

8.2 Limitations

- **$T_{\text{sky}}(r)$ is reconstructed.** Stark’s map comes from two-dimensional PSF simulations that are not published, and this model rescales the azimuthally averaged core-throughput curve instead. The term matters: with a constant T_{sky} , the mean realized yield falls from 22.4 to 19.2 (Table 5). A different reconstruction would shift the absolute yields, though hardly the yield ratio.
- **The coronagraph is azimuthally averaged.** The digitized one-dimensional DMVC6 curves stand in for two-dimensional maps.
- **Exozodi draws are interpolated.** The 200 draws (Stark uses 500) are evaluated on a 23-level grid, and the interpolation changes optimized yields by at most 0.24%. Stark also draws the HOSTS distribution itself from its posterior; he finds the effect negligible, and it is not modeled here.
- **The $\eta_{\oplus}$ distribution is reconstructed and departs from Stark’s text** (Sec. 3.2). It drives 99.1% of the remaining disagreement with Fig. 10 but barely moves the yield ratio (Table 5). The 86% reading and the Bryson-mixture law are available as options.
- **α and β are dominated by their priors.** The ratio ρ_B inherits the shape of SAG13 rather than coming from a new fit to Kepler DR25.
- **There is no stray light from binary companions.** AYO includes it; for HIP 77052 it is about 10% of the background.
- **The area fill factor (0.785) has no published source.** It moves the yield by about $\pm 4\%$.

9 Conclusions

1. For a mass-selected planet sample ($0.5\text{--}2 M_{\oplus}$) in a narrowed habitable zone ($0.96\text{--}1.20$ AU), the baseline 6 m HWO mission would find a median of 4.3 candidates, against 18.8 for the canonical selection. The yield ratio is $0.232^{+0.027}_{-0.021}$ (68% interval), larger than the occurrence ratio because the survey reallocates the freed characterization time and the narrower zone drops the faintest planets.
2. The probability of finding at least 25 candidates falls from 35.1% to 1.20%, which puts the decadal benchmark out of reach at this aperture.
3. The yield ratio stays between 0.2201 and 0.2386 across seven model variants spanning both readings of the $\eta_{\oplus}$ interval, both albedo methods, the presence or absence of the T_{sky} term, and a different functional form for $\eta_{\oplus}$.
4. The reimplementaion, built from the published description with nothing fitted to Stark’s results, agrees with 21 of 24 independent published comparisons and reproduces the shape of the published yield distribution.
5. Characterization times depend more steeply on target difficulty and aperture than in AYO, and no throughput setting removes the difference. The reading of the $\eta_{\oplus}$ interval accounts for 99.1% of the remaining disagreement with the published yield distribution. Stark’s per-visit albedo test gives a penalty of 3.6% against his reported 12%.

Data and code availability

The analysis is a fifteen-step vaibify pipeline in `/workspace/yields`. Each step declares its inputs and outputs and carries integrity, qualitative and quantitative tests generated from those outputs, and the physics library has 91 unit tests. Every number in this report is inserted at build time from the output of a pipeline step (`cd doc && python3 buildReport.py`). Table 6 maps the steps onto the graphical model.

Step	Name	Role
A01	buildTargetCatalog	$\mathbf{s}_i$: HPIC, 11,757 stars
A02	modelCoronagraph	$\mathcal{M}, \mathcal{B}, p_z$ (digitized Fig. 9), pinned z_i, p_A and the albedo test
A03	calibrateToStark	the expected 6 m yield with nothing fitted, and the diagnostic throughput factor
A04	computeCompleteness	$\mathbf{o}_{ij}, A_{ij}, \tau_{ij}, C_i$ on a grid of z ; 200 draws of $\mathbf{z}$
A05	optimizeSurvey	$\lambda, \mathbf{a}_i$ at $\eta = 0.24$, at fixed z and per draw
A06	sampleOccurrencePosterior	$\eta_{\oplus}, \alpha, \beta, \rho_{\mathcal{B}}$
A07	predictRedefinedYield	$\mathbf{a}_i(\eta)$ on a grid, Y by Eq. (3)
A08	compareApertureScaling	the same at 6–9 m, over draws of $\mathbf{z}$
A09	verifyAgainstPublished	Table 2
A10	compareTargetCompleteness	per-target comparison with Fig. 11
A11	tabulateEarthZones	Sec. 5.2, Appendix C
A12	comparePublishedFigures	Appendix A
A13	calibrationDegeneracy	the throughput-parameter design of Sec. 7
A14	compareModelVariants	the variants and the 2×2 decomposition of Sec. 6
A15	testOccurrenceIntervalReading	the test of the $\eta_{\oplus}$ interval against its source (Sec. 3.2)

Table 6: The pipeline steps and their role in the model.

A Published figures beside their model counterparts

Each original is copied unaltered from the arXiv source of Stark et al. (2024) (arXiv:2405.19418, ms_v2). Step A12 draws the model versions with the published axes, ranges and line styles. Figures with no like-for-like counterpart (optical layouts, throughput and quantum-efficiency curves, and the design-change scenarios of Stark’s Sec. 5) are omitted.

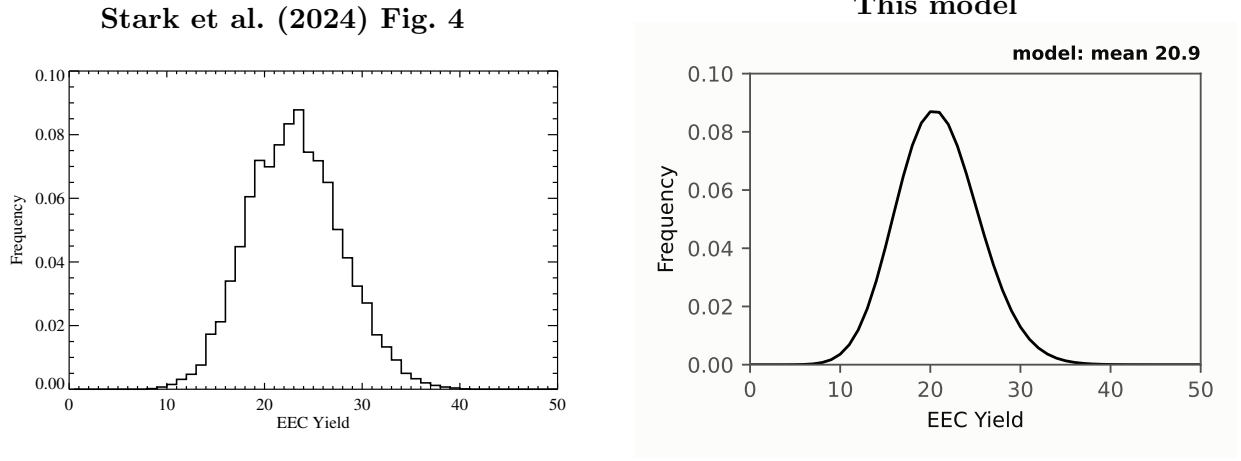


Figure 7: Planet sampling as the only source of uncertainty: $A_G = 0.2$, exozodi fixed at 3 zodis, and Poisson counting of planets. Published: mean 22.5, standard deviation 21% of the mean. Model: mean 20.9, with nothing fitted to the published yield.

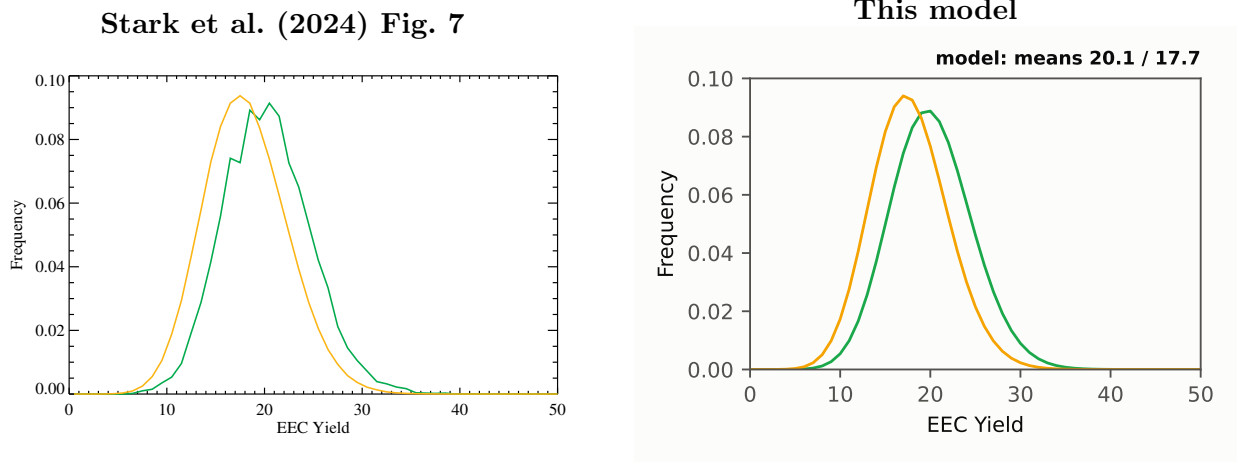


Figure 8: Yield with albedo drawn and exozodi fixed (green), and with exozodi drawn as well (orange), both at fixed $\eta_{\oplus}$. Published means: 19.8 and 17.6. Model means: 20.1 and 17.7, over 200 draws.

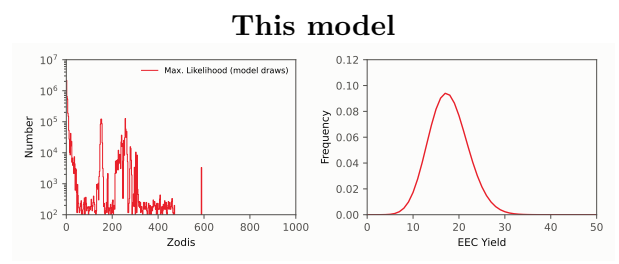
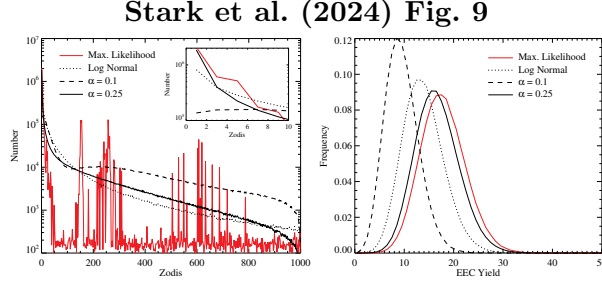


Figure 9: Left: exozodi levels drawn from the LBTI HOSTS maximum-likelihood distribution (red). The model panel is a histogram of the levels the pipeline actually drew, scaled to Stark’s 500×10^4 draws; its median is 3.00 zodis. Stark’s other three distributions are not modeled. Right: the yield those levels produce at fixed $\eta_{\oplus}$ (red).

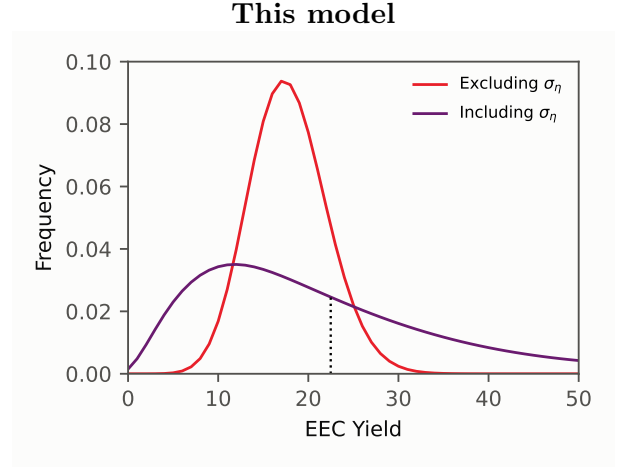
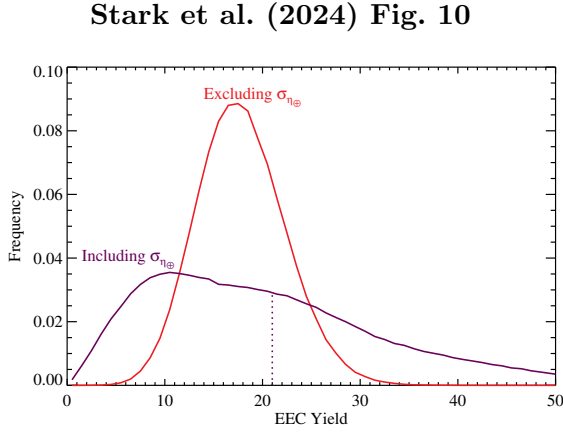


Figure 10: Realized yield including (purple) and excluding (red) the uncertainty in $\eta_{\oplus}$; the dotted line marks the mean of the purple curve. Total-variation distance between model and published curves: 0.027 (purple) and 0.052 (red). Means (purple / red): published 21.0 / 17.3, model 22.5 / 17.8.

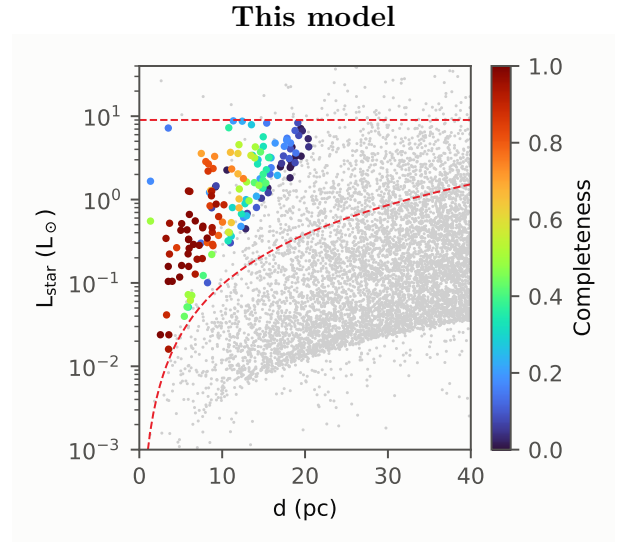
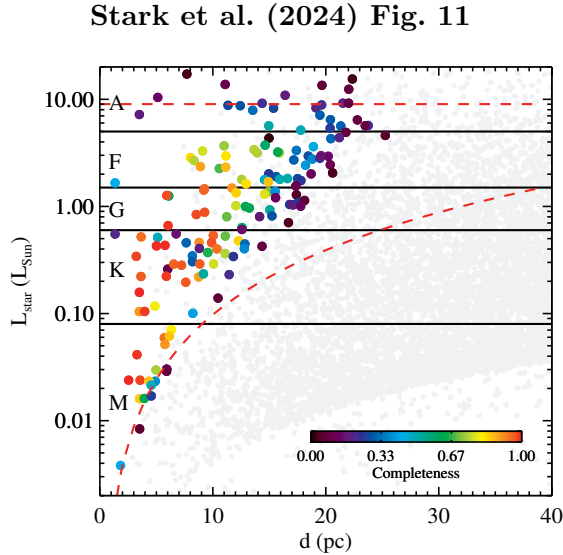
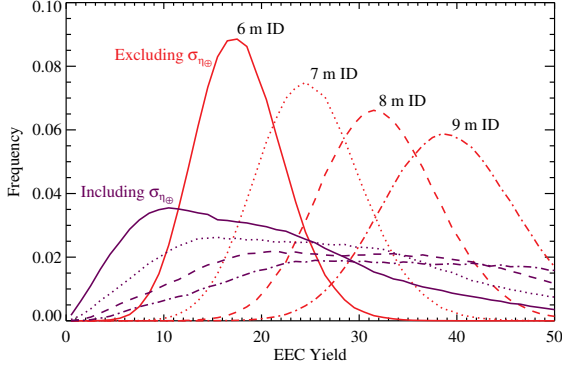


Figure 11: Targets selected in one representative exozodi draw, colored by completeness and plotted over the full input list (gray). Published: 154 targets, summed completeness 77.4, median 0.50. Model (draw 14, the draw with the median yield): 157 targets, summed completeness 77.0, and median 0.09 over the 141 targets that match. The red dashed lines are the noise floor for a $1.4 R_{\oplus}$ planet at the EEID, and the outer edge of the habitable zone at $1.5 \lambda/D$ at $1 \mu\text{m}$. The spectral-type lines are not redrawn.

Stark et al. (2024) Fig. 12



This model

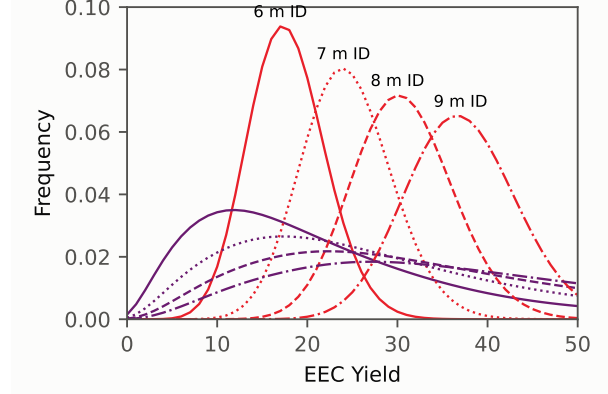
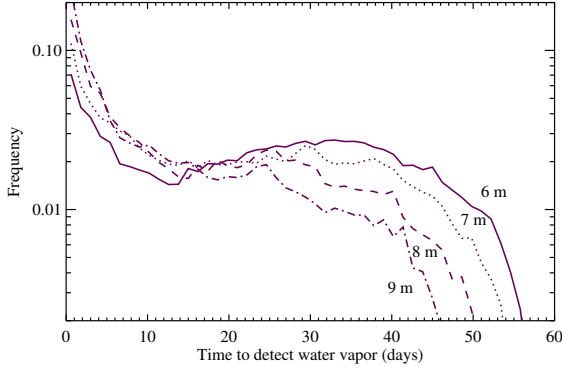


Figure 12: Realized yield at inscribed diameters of 6, 7, 8 and 9 m (solid, dotted, dashed and dot-dashed), including (purple) and excluding (red) the uncertainty in $\eta_{\oplus}$, with no calibration at any aperture. Excluding that uncertainty, Stark’s 9 m yield is 2.2 times his 6 m yield; the model’s is 2.09 times.

Stark et al. (2024) Fig. 14



This model

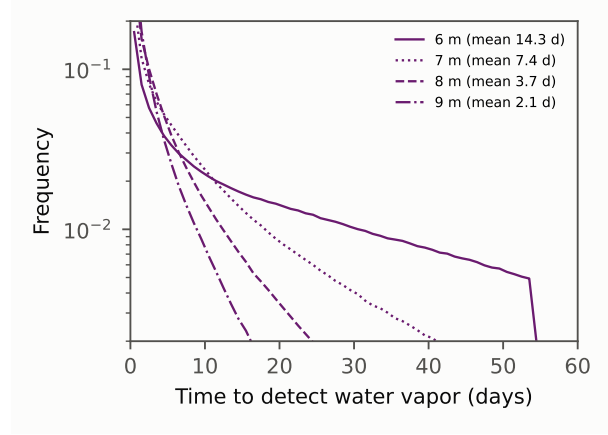
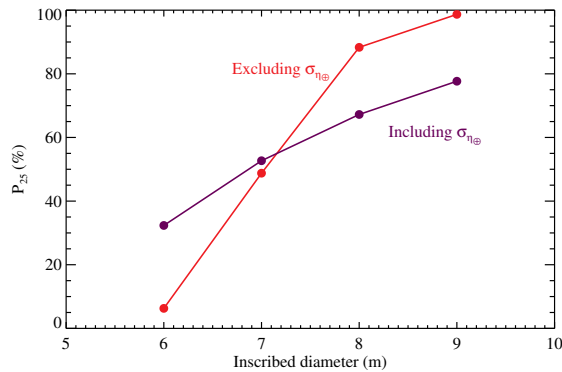


Figure 13: Characterization times of the individual EECs among the first 18, at 6–9 m. The model uses 20 exozodi draws, recomputed planet by planet, in 1-day bins. Published means: 22 days at 6 m and 3.5 days at 9 m. Model means at 6/7/8/9 m: 14.3 / 7.4 / 3.7 / 2.1 days. Stark does not state how his histograms are normalized, and with 1-day bins his 9 m plateau alone would exceed his stated mean, so compare the shapes rather than the heights. His distributions keep a plateau of 15–50-day spectra at every aperture; the model’s do not.

Stark et al. (2024) Fig. 15



This model

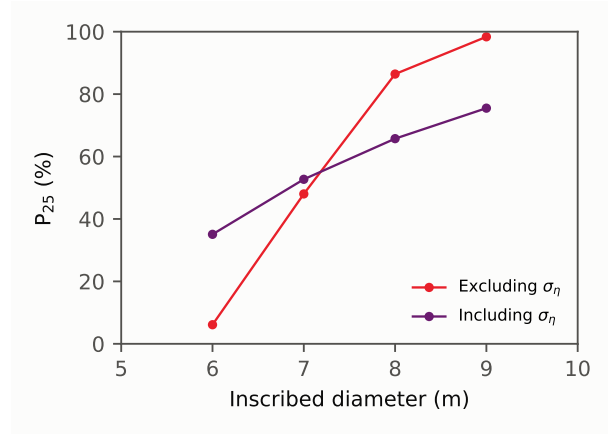
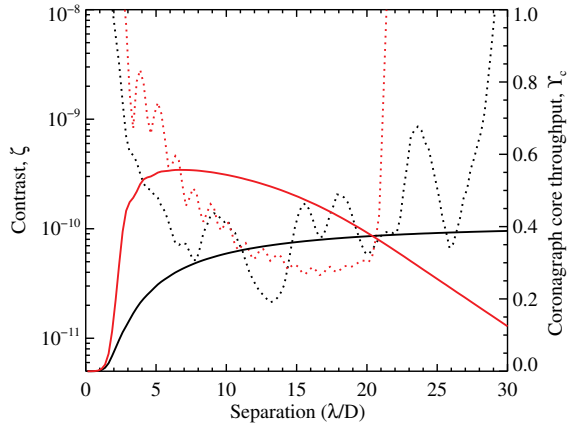


Figure 14: P_{25} , the probability of at least 25 candidates, against inscribed diameter (6/7/8/9 m). Including the uncertainty in $\eta_{\oplus}$: published 32/53/67/78%, model 35 / 53 / 66 / 76%. Excluding it: published 6/49/89/99.5%, model 6 / 48 / 86 / 98%.

Stark et al. (2024) Fig. 25



This model

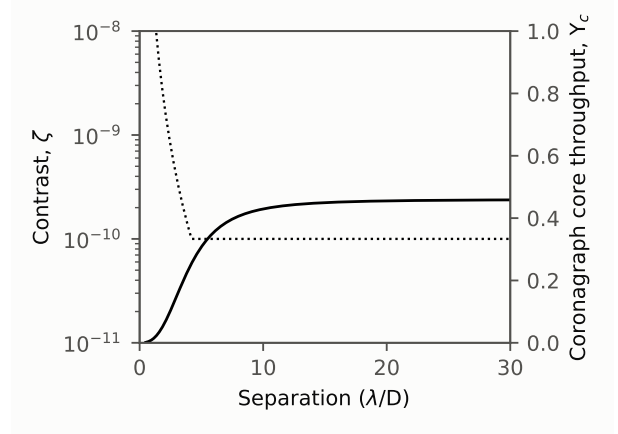


Figure 15: DMVC6 raw contrast (dotted) and core throughput (solid) against separation in circumscribed λ/D . The model curves are the digitized published ones, so this pair checks the digitization, not the model. The model's contrast is drawn with the 10^{-10} floor applied, as the exposure-time calculation uses it; the published curve is the raw simulation. The published figure also shows the PIAA-FPM2.5 coronagraph (red), which is not used here.

B Implementation choices and their effect

Each row states a modeling choice, its source where one is published, and what the alternative does in this model.

Choice	Source	Effect of the alternative
Up to six visits per star	Stark et al. (2015)	A single visit lowers the yield by a factor of about 1.7.
Characterization budget is the mean time over counted planets	—	A median over all detectable planets flattens the aperture scaling from $D^{1.88}$ to $D^{1.11}$.
Spectra exceeding the cap are not charged	—	Charging them costs up to 15 000 days at 20 pc for observations never attempted.
Target list is HPIC	Stark et al. (2024)	A Gaia-based list misses 26–34% of distant FGK dwarfs and is biased toward hotter stars.
Coronagraph curves in circumscribed λ/D	Stark et al. (2024), Sec. 6.1	Inscribed λ/D puts the IWA 19% too far out and lowers the yield by about 20%.
Collecting area is the full obscured primary	Stark et al. (2019), Sec. 6.2	An inscribed circle counts light lost at the Lyot stop twice.
Spectra at the most favorable orbital phase	Stark et al. (2019), Sec. 5	Detection-phase spectra lengthen characterization times.
Characterization budgeted at $A_G = 0.2$	Stark et al. (2024), Sec. 3.2	Budgeting at the drawn albedo lowers the albedo penalty.
Albedo counted by the per-visit flux test, both bounds	Stark et al. (2024), Sec. 3.2	Re-derived exposures give an albedo penalty of 8.6% instead of 3.6% (Table 5).
T_{sky} depends on separation	Stark (2019), Eqs. 5–6	A constant value lowers the mean realized yield to 19.2.
Exozodi from the HOSTS distribution, re-optimized per draw	Stark et al. (2024), Sec. 3.3	A lognormal stand-in puts 0.2% of stars above 100 zodis (HOSTS: 21%) and gives a 0.7% exozodi penalty.
Four stars pinned to LBTI values	Stark et al. (2024), Sec. 3.3	Worth at most 0.47 EEC.
Zodi from Leinert, by ecliptic latitude	Stark et al. (2014)	A constant zodi departs from AYO’s benchmark by factors of 0.56–1.26.
Noise floor in the exposure-time equation; overheads inside the cap	Stark et al. (2019, 2024)	Neither changes the characterization-time discrepancy.
No throughput calibration	—	The factor that would reproduce 22.5 is 1.20; it would absorb uniform errors in exposure time.
$\eta_{\oplus}$ interval read as 68%	Stark et al. (2024), Fig. 10	The stated 86% raises the distance to Fig. 10 from 0.037 to 0.131.
Equal-slope allocation	Stark et al. (2014)	Within 0.00% of an exact optimum.
Blackbody stellar flux	—	Agrees with measured V to +0.019 mag.
One detection before characterization	Stark et al. (2024)	Requiring two lowers the yield by about 8% and does not narrow its distribution.

Table 7: Implementation choices. The explanations tested and rejected for the characterization-time discrepancy are listed in Sec. 7.

C Every contributing star

This table lists every star with an expected yield of at least 0.01 EEC in either zone's survey, ordered by its classic-zone yield. Zone edges are angular separations in milliarcseconds; C is the completeness reached, averaged over exozodi draws; and N is the expected number of candidates. The complete table, covering every screened star with every column (AU, λ/D at 550 and 1000 nm, time allocated, characterization time, and the fraction of draws in which the star is observed), is `TabulateEarthZones/earthZoneTable.csv`.

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
alf Cen B	K1V	1.35	0.552	529–661	524–921	0.49	0.65	0.250	0.052	0.21
HD 95735	M2+V	2.55	0.0239	58–73	58–101	0.99	1.00	0.249	0.052	0.21
tau Cet	G8V	3.65	0.52	189–237	188–330	0.98	1.00	0.248	0.052	0.21
eps Ind	K5V	3.64	0.222	124–155	123–216	0.98	1.00	0.247	0.052	0.21
61 Cyg A	K5V	3.50	0.158	109–136	108–190	0.98	1.00	0.247	0.052	0.21
61 Cyg B	K7V	3.50	0.104	89–111	88–154	0.98	1.00	0.246	0.052	0.21
HD 217987	M2V	3.29	0.0413	59–74	59–103	0.97	1.00	0.245	0.052	0.21
V* AX Mic	M1V	3.97	0.105	78–98	77–136	0.96	1.00	0.239	0.051	0.21
70 Oph A	K0V	5.11	0.514	135–168	133–234	0.95	1.00	0.237	0.051	0.22
del Equ	F7(V)+G	4.51	0.265	110–137	108–191	0.95	1.00	0.236	0.052	0.22
omi02 Eri	K0V	5.01	0.428	125–157	124–218	0.95	1.00	0.236	0.052	0.22
eps Eri	K2V	3.22	0.342	174–218	173–303	0.96	1.00	0.236	0.051	0.22
70 Oph B	K4V	5.10	0.162	76–95	75–132	0.92	0.99	0.229	0.051	0.22
e Eri	G6V	6.04	0.663	129–162	128–225	0.91	0.99	0.223	0.051	0.23
36 Oph	K2V+K1V	5.95	0.334	93–117	92–162	0.90	0.97	0.222	0.050	0.22
sig Dra	K0V	5.76	0.433	110–137	108–191	0.89	0.98	0.221	0.051	0.23
IRAS 20079-3614		6.01	0.26	81–102	81–142	0.88	0.94	0.221	0.049	0.22
HD 88230	K6VeFe-	4.87	0.117	68–84	67–118	0.89	0.97	0.220	0.050	0.23
HD 131977	K4V	5.89	0.223	77–96	76–134	0.87	0.94	0.218	0.049	0.22
V* GX And	M2V	3.56	0.0239	42–52	41–72	0.86	0.88	0.217	0.046	0.21
36 Oph B	K1V	5.95	0.331	93–116	92–162	0.87	0.96	0.216	0.049	0.23
eta Cas	F9V	5.92	1.27	182–228	181–317	0.88	0.99	0.214	0.051	0.24
ksi Boo	G7Ve+K5	6.75	0.554	106–132	105–184	0.86	0.92	0.213	0.048	0.22
HD 219134	K3V	6.54	0.29	79–99	78–137	0.84	0.89	0.211	0.046	0.22
del Pav	G8IV	6.10	1.25	176–220	174–306	0.87	0.99	0.210	0.050	0.24
V* V2215 Oph	K5V	5.95	0.158	64–80	63–111	0.84	0.87	0.210	0.045	0.22
107 Psc	K1V	7.64	0.458	85–106	84–148	0.83	0.87	0.205	0.045	0.22
mu. Cas	G5Vb	7.68	0.479	87–108	86–151	0.79	0.82	0.194	0.042	0.22
HD 173739	M3V	3.52	0.016	35–43	34–60	0.76	0.69	0.193	0.036	0.19
61 Vir	G6.5V	8.53	0.841	103–129	102–179	0.78	0.84	0.192	0.043	0.22
HD 16160	K3V	7.23	0.282	71–88	70–123	0.77	0.80	0.191	0.041	0.22
bet CVn	G0V	8.47	1.22	125–156	124–218	0.78	0.85	0.190	0.043	0.23
HD 156384	K3V+K5V	6.97	0.192	60–76	60–105	0.74	0.77	0.186	0.040	0.21
HD 4628	K2.5V	7.43	0.3	71–88	70–123	0.73	0.75	0.181	0.039	0.22
V* TW PsA	K4Ve	7.60	0.196	56–70	55–97	0.72	0.78	0.180	0.041	0.23
p Eri A	K2V	8.19	0.345	69–86	68–120	0.72	0.77	0.179	0.040	0.22
chi01 Ori	G0V	8.70	1.11	116–145	115–202	0.73	0.79	0.177	0.041	0.23
p Eri B	K2V	8.20	0.305	65–81	64–112	0.71	0.76	0.177	0.039	0.22
ksi Boo B	K5Ve	6.75	0.127	51–63	50–88	0.70	0.74	0.177	0.038	0.22
HD 192310	K2+V	8.81	0.405	69–87	69–121	0.71	0.79	0.175	0.041	0.23
ksi UMa	F8.5:V+	8.73	0.796	98–123	97–171	0.71	0.77	0.175	0.039	0.22
ksi UMa A	F8.5:V	8.73	1.26	123–154	122–214	0.72	0.80	0.173	0.040	0.23
zet Tuc	F9.5V	8.61	1.28	126–158	125–220	0.70	0.76	0.171	0.039	0.23
41 Ara	G9V	8.79	0.461	74–93	73–129	0.69	0.75	0.169	0.039	0.23
HD 131976	M1.5V	5.90	0.0723	44–55	43–76	0.67	0.67	0.169	0.035	0.21
kap01 Cet	G5V	9.27	0.881	97–122	96–169	0.69	0.76	0.168	0.039	0.23
HD 102365	G2V	9.32	0.849	95–119	94–165	0.68	0.77	0.166	0.040	0.24
bet Com	F9.5V	9.20	1.42	124–155	123–216	0.69	0.76	0.165	0.039	0.23
alf Cen A	G2V	1.35	1.66	918–1147	908–1596	0.18	0.24	0.164	0.030	0.18
61 UMa	G8V	9.57	0.625	79–99	78–138	0.67	0.76	0.163	0.039	0.24
chi Dra	F7V	8.30	2.19	171–214	170–298	0.69	0.83	0.163	0.041	0.25
gam Pav	F9VFe-1	9.26	1.46	125–157	124–218	0.67	0.76	0.160	0.039	0.24
pi.03 Ori	F6V	8.02	2.85	202–253	200–352	0.68	0.86	0.157	0.043	0.27
mu.01 Her	G5IV	8.33	2.68	188–235	186–328	0.67	0.86	0.156	0.042	0.27
alf Men	G7V	10.21	0.866	87–109	87–152	0.64	0.74	0.154	0.038	0.25
gam Lep	F6V	8.90	2.36	166–207	164–288	0.66	0.79	0.154	0.040	0.26
bet Hyi	G0V	7.48	3.57	243–303	240–422	0.64	0.92	0.152	0.044	0.29
del Tri	G0.5VFe	10.91	1.21	97–121	96–168	0.63	0.80	0.152	0.041	0.27
HD 10780	K0V	10.04	0.536	70–88	69–122	0.60	0.73	0.147	0.038	0.26
HD 32147	K3+V	8.84	0.29	58–73	58–102	0.59	0.67	0.146	0.035	0.24
HD 79210	K7V	6.33	0.0813	43–54	43–75	0.57	0.52	0.144	0.027	0.19

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
tet Per	F8V	11.14	2.32	131–164	130–228	0.61	0.82	0.142	0.041	0.29
12 Oph	K1V	9.89	0.46	66–82	65–114	0.58	0.70	0.142	0.036	0.26
iot Per	G0V	10.57	2.26	136–171	135–237	0.61	0.78	0.141	0.039	0.28
V* AK Lep	K3	8.89	0.279	57–71	56–99	0.56	0.63	0.139	0.033	0.24
del Eri	K0+IV	9.09	3.29	192–240	190–333	0.59	0.82	0.136	0.040	0.29
HD 36395	M1.5Ve	5.70	0.0594	41–51	41–71	0.53	0.47	0.134	0.024	0.18
11 LMi	G8Va	11.23	0.801	77–96	76–133	0.53	0.70	0.128	0.036	0.28
zet Dor	F9VFe-0	11.69	1.48	100–125	99–174	0.53	0.72	0.125	0.037	0.29
lam Ser	G0-V	11.90	2.03	115–144	114–200	0.54	0.73	0.125	0.037	0.30
zet TrA	F9V	12.07	1.34	92–115	91–160	0.51	0.71	0.121	0.036	0.30
20 Crt	K0V	9.56	0.371	61–77	61–106	0.48	0.57	0.120	0.029	0.24
lam Aur	G1.5IV-	12.56	1.78	102–128	101–178	0.50	0.72	0.117	0.036	0.31
gam Ser	F6V	11.16	2.96	148–185	147–258	0.50	0.71	0.114	0.034	0.30
HD 50281	K3.5V	8.74	0.219	51–64	51–89	0.46	0.41	0.114	0.021	0.19
HD 42581	M1V	5.76	0.0515	38–47	37–66	0.44	0.19	0.111	0.010	0.09
HD 79211	M0V	6.33	0.0707	40–50	40–70	0.44	0.24	0.111	0.013	0.11
zet02 Ret	G1V	12.04	1.02	80–100	80–140	0.46	0.63	0.110	0.032	0.29
bet Vir	F9V	11.00	3.69	168–209	166–292	0.48	0.74	0.108	0.035	0.33
iot Peg	F5V	11.90	3.57	152–191	151–265	0.48	0.75	0.108	0.035	0.33
36 UMa	F8V	12.94	1.63	95–118	94–165	0.46	0.66	0.107	0.034	0.31
zet01 Ret	G2.5VHd	12.04	0.795	71–89	70–124	0.44	0.53	0.106	0.027	0.26
HD 17925	K1V	10.36	0.401	59–73	58–102	0.41	0.38	0.100	0.019	0.19
HD 191849	M0V	6.16	0.0616	39–48	38–67	0.38	0.15	0.095	0.008	0.08
54 Psc	K0.5V	11.10	0.529	63–79	62–109	0.38	0.39	0.092	0.020	0.22
ups And	F9V	13.47	3.38	131–164	130–228	0.41	0.67	0.090	0.032	0.35
HD 157881	K7V	7.71	0.123	44–55	43–76	0.35	0.16	0.087	0.008	0.09
HD 103095	K1V_Fe-	9.17	0.235	51–63	50–88	0.34	0.23	0.085	0.012	0.14
10 Tau	F9IV-V	13.92	3.16	123–153	121–213	0.37	0.61	0.084	0.029	0.35
47 UMa	G1-VFe-	13.88	1.6	87–109	86–152	0.36	0.50	0.084	0.025	0.30
gam Vir	F1-F2V	12.02	4.55	170–213	169–296	0.38	0.65	0.083	0.029	0.36
alf CMi	F5IV-V+	3.50	7.21	737–921	729–1282	0.15	0.30	0.080	0.028	0.36
tau01 Eri	F7V	14.27	2.67	110–137	109–191	0.34	0.53	0.080	0.026	0.33
iot Psc	F7V	13.66	3.39	129–162	128–225	0.35	0.59	0.079	0.028	0.36
HD 119850	M2V	5.43	0.0397	35–44	35–61	0.30	0.06	0.076	0.003	0.04
alf For	F6V	14.00	4.28	142–177	140–247	0.35	0.60	0.076	0.027	0.36
HD 104304	G8IV	12.76	0.892	71–89	70–124	0.32	0.34	0.076	0.017	0.23
HD 223778	K3V	10.89	0.399	56–70	55–97	0.30	0.19	0.074	0.010	0.13
eta Cas B	K7Ve	5.93	0.0513	37–46	36–64	0.29	0.04	0.074	0.002	0.03
alf Com	F5V+F6V	14.32	3	116–145	115–202	0.32	0.51	0.072	0.025	0.35
HD 147513	G5V	12.89	0.998	74–93	74–129	0.29	0.35	0.071	0.018	0.25
V* DE Boo	K0.5V	11.63	0.541	61–76	60–106	0.29	0.21	0.070	0.011	0.15
HD 13445	K1.5V	10.76	0.406	57–71	56–99	0.28	0.20	0.070	0.010	0.14
gam Vir B	F0mF2V	12.74	5.27	173–216	171–301	0.32	0.56	0.069	0.025	0.37
HD 122064	K3V	10.06	0.291	51–64	51–89	0.28	0.12	0.068	0.006	0.09
58 Eri	G2.5IV-	13.23	0.966	71–89	71–124	0.28	0.26	0.066	0.013	0.20
psi Cap	F5V	14.63	3.72	127–158	125–220	0.29	0.48	0.064	0.023	0.35
85 Peg	G5VbFe-	12.32	0.674	64–80	63–111	0.26	0.21	0.064	0.011	0.17
111 Tau	F8V	14.57	1.78	88–110	87–153	0.27	0.35	0.062	0.017	0.28
HD 10307	G1V	14.67	1.89	90–112	89–157	0.26	0.37	0.061	0.019	0.31
26 Dra	F9V+K3V	14.43	1.48	81–101	80–141	0.26	0.31	0.060	0.016	0.26
HD 211415	G0V	14.06	1.17	74–92	73–129	0.25	0.23	0.059	0.012	0.20
chi Her	F8VFe-2	15.90	3.16	107–134	106–187	0.26	0.40	0.058	0.020	0.34
zet Her	G0IV	10.80	7.26	240–300	237–417	0.26	0.49	0.055	0.020	0.36
nu. Phe	F9VFe+0	15.25	2.03	90–112	89–156	0.24	0.30	0.055	0.015	0.28
b Aql	G7IVHde	14.91	1.7	84–105	83–146	0.23	0.28	0.053	0.014	0.27
alf Crv	F1V	14.97	4.34	134–167	132–232	0.23	0.39	0.051	0.018	0.35
HD 84117	F9V	14.95	2.01	91–114	90–159	0.22	0.30	0.051	0.015	0.29
h Dra	F8V	15.33	2.28	95–118	94–164	0.22	0.29	0.051	0.015	0.29
tau Boo	F7IV-V	15.60	3.18	110–137	109–191	0.23	0.33	0.050	0.016	0.32
sig Boo	F4VkF2m	15.75	3.22	109–137	108–190	0.21	0.29	0.047	0.014	0.30
eta Lep	F2V	14.95	5.66	153–191	151–266	0.22	0.37	0.047	0.016	0.35
171 Pup	F9V	15.27	1.57	79–99	78–137	0.19	0.19	0.046	0.010	0.21
m Tau	G4V	15.91	2.41	94–117	93–163	0.20	0.27	0.045	0.013	0.29
18 Sco	G2Va	14.13	1.1	71–89	70–124	0.19	0.14	0.045	0.007	0.16
mu. Ara	G3IV-V	15.60	1.9	85–106	84–148	0.20	0.22	0.044	0.011	0.25
20 LMi	G3VaHde	14.92	1.37	75–94	75–131	0.18	0.15	0.043	0.008	0.18
b Her	F6V	15.76	2.02	87–108	86–151	0.19	0.21	0.043	0.010	0.24
HD 176051	F9V+K1V	14.89	1.42	77–96	76–134	0.18	0.16	0.043	0.008	0.19
HD 69830	G8:V	12.58	0.609	60–74	59–104	0.18	0.05	0.043	0.002	0.06
HD 160346	K3-V	10.71	0.32	51–63	50–88	0.17	0.02	0.042	0.001	0.03
ksi Peg	F6V	16.39	4.76	128–160	126–222	0.19	0.30	0.042	0.013	0.32
rho01 Cnc	K0IV-V	12.58	0.635	61–76	60–106	0.17	0.06	0.041	0.003	0.07
HD 166620	K2V	11.09	0.363	52–65	52–91	0.16	0.02	0.040	0.001	0.03
tet UMa	F7V	13.53	7.9	199–249	197–347	0.18	0.35	0.037	0.013	0.36

Star	SpT	d (pc)	L ($L_{\odot}$)	Earth zone (mas)	classic HZ (mas)	C_{HZ}	C_{EZ}	N_{HZ}	N_{EZ}	ratio
eta Boo	G0IV	11.34	8.79	251–314	248–437	0.16	0.33	0.035	0.013	0.37
bet TrA	F1V	12.42	8.71	228–285	226–397	0.16	0.30	0.033	0.012	0.35
I Car	F3V	16.28	5.25	135–169	134–235	0.14	0.22	0.031	0.010	0.33
HD 172051	G6V	13.02	0.682	61–76	60–106	0.12	0.02	0.029	0.001	0.03
i Boo	G0Vn	12.95	0.639	59–74	59–103	0.12	0.01	0.028	0.000	0.02
gam CrA A	F8V	17.03	2.47	89–111	88–154	0.13	0.12	0.028	0.006	0.20
tau01 Hya	F5V	18.24	4.05	106–132	105–184	0.13	0.19	0.028	0.008	0.30
HD 3443	K1V+G	15.53	1.32	71–89	70–123	0.12	0.05	0.027	0.002	0.08
9 Pup	G0V	16.67	1.95	80–101	80–140	0.12	0.07	0.026	0.004	0.14
HD 188088	K3VaCN1	14.14	0.803	61–76	60–106	0.11	0.01	0.026	0.000	0.02
nu.02 Lup	G3/5V	14.74	1.04	66–83	66–115	0.11	0.02	0.026	0.001	0.03
pi.01 UMa	G0.5V	14.43	0.972	66–82	65–114	0.11	0.01	0.026	0.001	0.03
phi02 Cet	F7V	15.92	1.79	81–101	80–140	0.11	0.07	0.025	0.004	0.15
gam CrA	F8V+F8V	17.89	4.8	118–147	116–204	0.11	0.13	0.023	0.006	0.26
HD 37394	K1	12.26	0.477	54–68	53–94	0.09	0.00	0.023	0.000	0.00
HD 115404	K2V	10.98	0.302	48–60	48–84	0.09	0.00	0.022	0.000	0.00
del Aql	F1IV-V(	15.37	8.26	179–224	178–312	0.11	0.20	0.022	0.008	0.35
70 Vir	G4Va	18.09	3.1	93–117	93–163	0.10	0.10	0.022	0.005	0.22
51 Peg	G2IV	15.51	1.38	73–91	72–126	0.09	0.03	0.021	0.002	0.07
g Lup	F3/5V	17.39	3.33	101–126	100–175	0.10	0.09	0.021	0.004	0.20
58 Oph	F5V	17.17	2.72	92–115	91–160	0.09	0.06	0.020	0.003	0.15
HD 74576	K3V	11.19	0.318	48–60	48–84	0.08	0.00	0.020	0.000	0.00
tau06 Eri	F5IV-V	17.78	5.15	123–153	121–213	0.09	0.11	0.020	0.005	0.26
psi05 Aur	F9V	16.61	1.82	78–97	77–136	0.08	0.02	0.019	0.001	0.06
ksi Oph	F2V	17.50	4.35	114–143	113–199	0.09	0.09	0.018	0.004	0.22
HD 103932	K4+V	10.17	0.212	43–54	43–76	0.07	0.00	0.018	0.000	0.00
HD 217357	K7+Vk	8.23	0.101	37–46	37–64	0.06	0.00	0.015	0.000	0.00
rho Gem	F1V	17.83	5.44	126–157	124–218	0.07	0.08	0.015	0.003	0.22
HD 5015	F9V	18.90	3.62	97–121	96–168	0.07	0.04	0.015	0.002	0.14
HD 207129	G2V	15.55	1.21	68–85	67–118	0.06	0.00	0.014	0.000	0.01
tet Cyg	F3+V	18.42	4.27	108–135	107–187	0.07	0.05	0.014	0.002	0.17
HD 114837	F6VFe-0	18.23	2.99	91–114	90–158	0.06	0.03	0.013	0.002	0.12
eta Crv	F2V	18.24	4.77	115–144	114–200	0.06	0.05	0.012	0.002	0.17
10 UMa	F3V+G5V	18.79	6.73	132–166	131–230	0.06	0.07	0.012	0.003	0.23
HD 32450	K7V	8.36	0.11	38–48	38–66	0.05	0.00	0.012	0.000	0.00
e Vir	G0V	17.57	2.19	81–101	80–141	0.05	0.01	0.010	0.000	0.03
HD 72673	K1V	12.16	0.403	50–63	50–87	0.04	0.00	0.010	0.000	0.00

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